

Linking and sharing multi-modal data with Knowledge Graphs

Application to intracranial aneurysms

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Séminaire AISTROSIGHT

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Comprendre les pathologies

cardio-vasculaires, métaboliques
et respiratoires pour mieux les
prévenir

Médecine translationnelle :

Patients $\leftrightarrow$ Recherche

Lien entre **fonction** biologique et
séquences **génétiques** ?

Bio-informatique

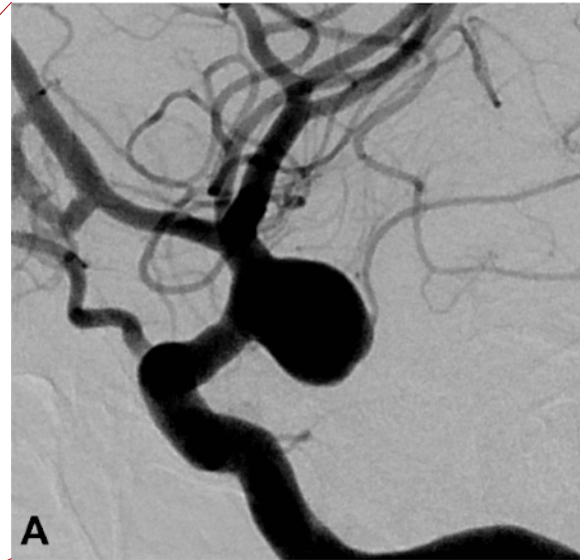
Données de santé en grand nombre

- **calcul haut-débit** : infrastructure distribuées
- **intégration de données** hétérogènes
- **modèles explicatifs / prédictifs (IA)**

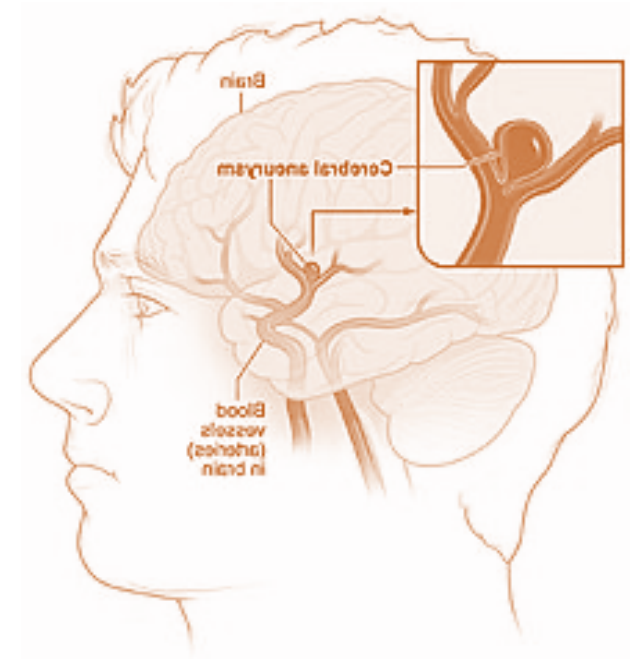
données massives $\rightarrow$ nouvelles connaissances

Intracranial aneurysms

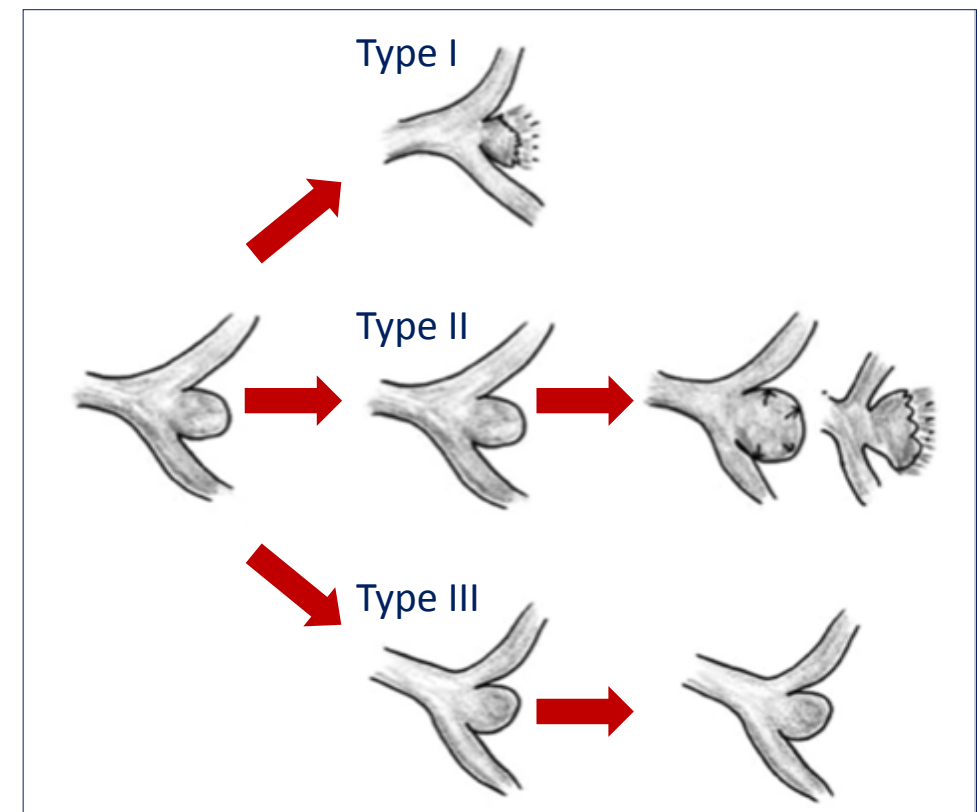
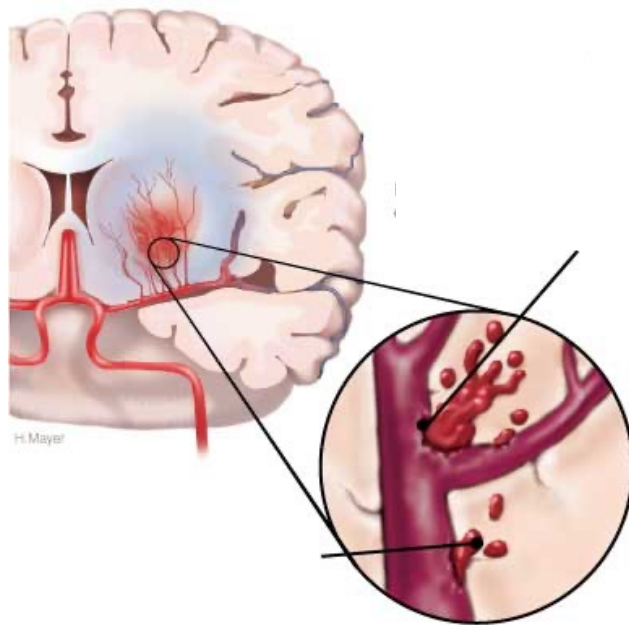
Intracranial aneurysm (ICA)



- ▶ 3% of the general population
- ▶ unpredictable rupture
- ▶ 50% of death in case of rupture



ICA rupture / stroke



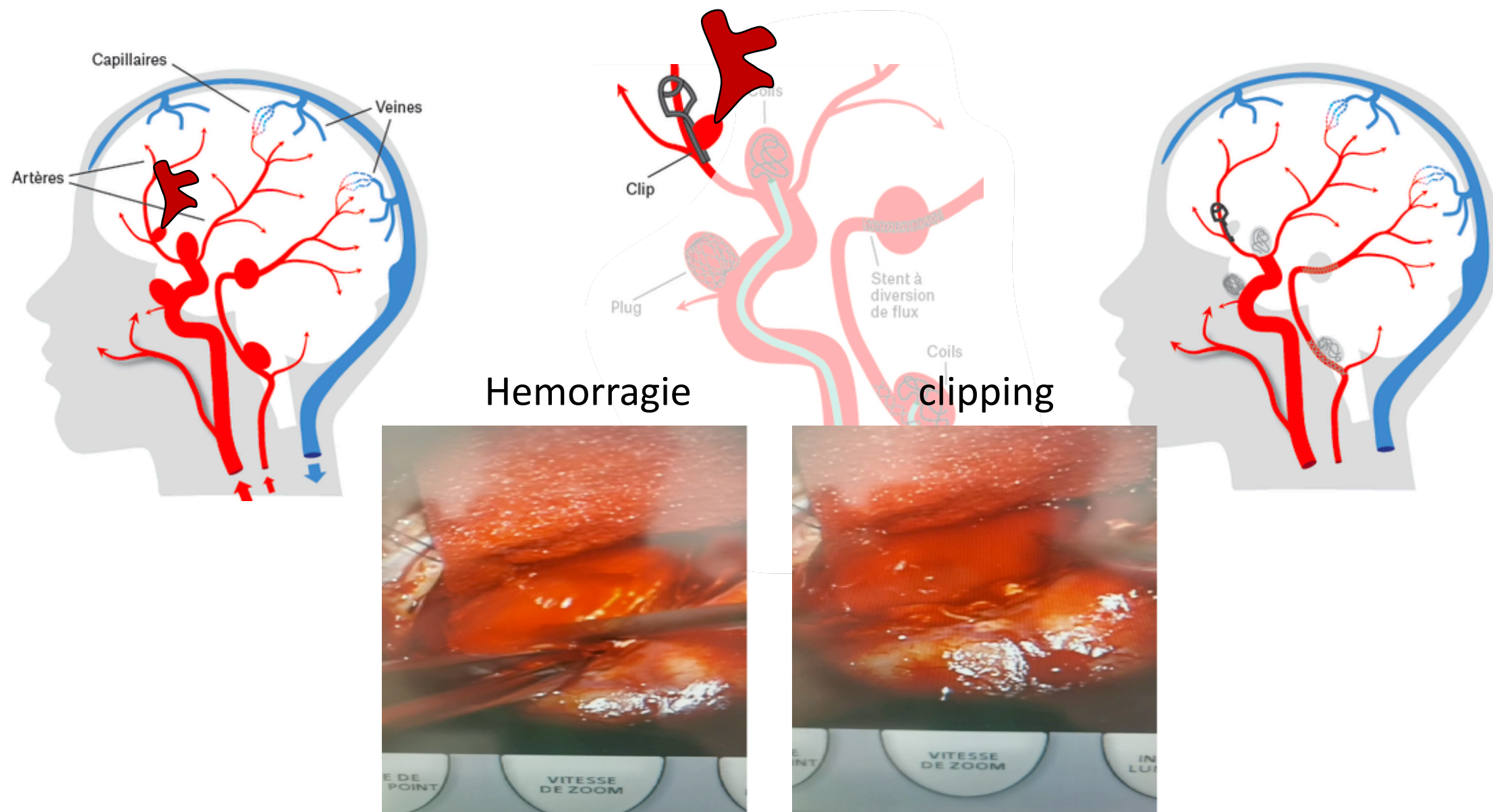
Risk factors

- ▶ smoking, alcohol consumption, hypertension, female sex, age
- ▶ familial history (RRx4)

In 10% of ICA cases = familial form of AIC with at least 2 first-degree relatives affected

How are they treated ?

Surgical clipping

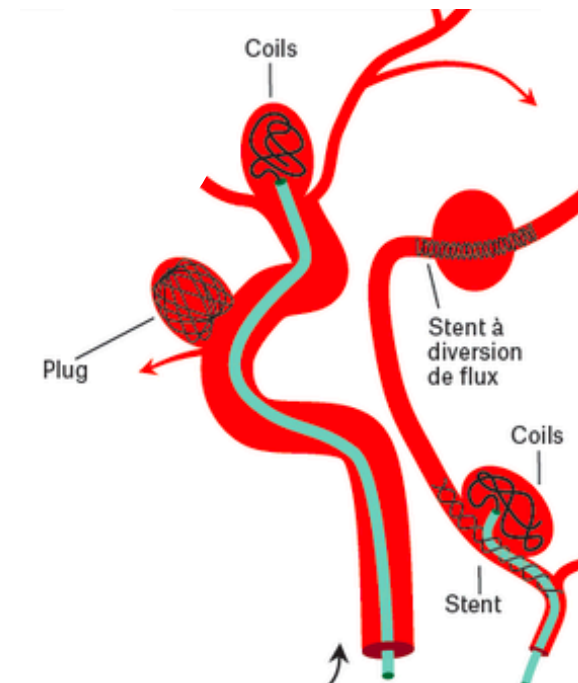
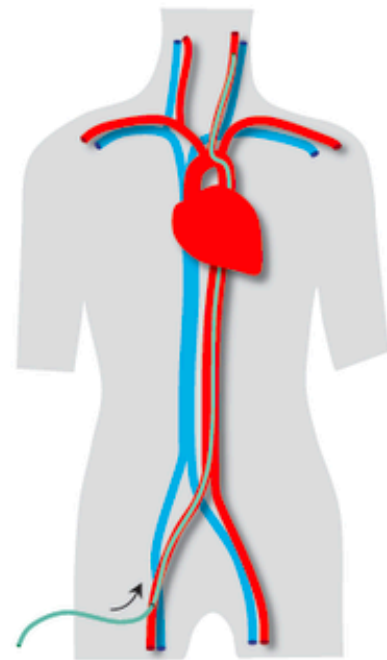
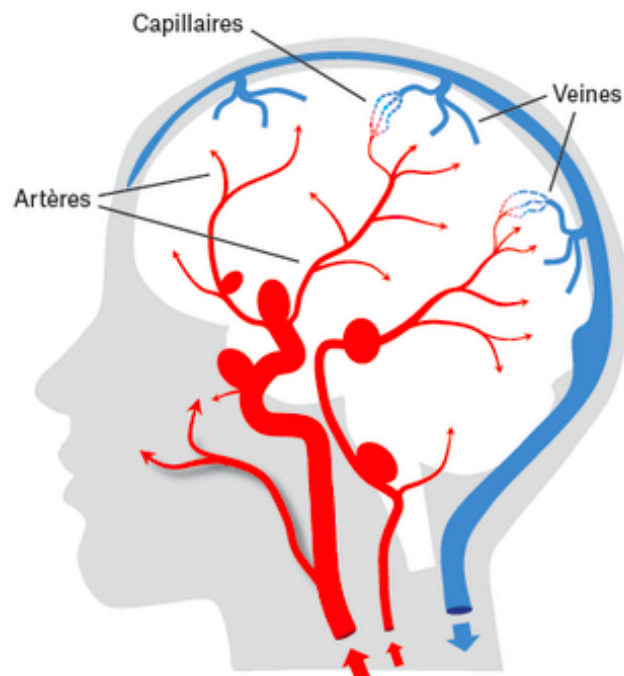


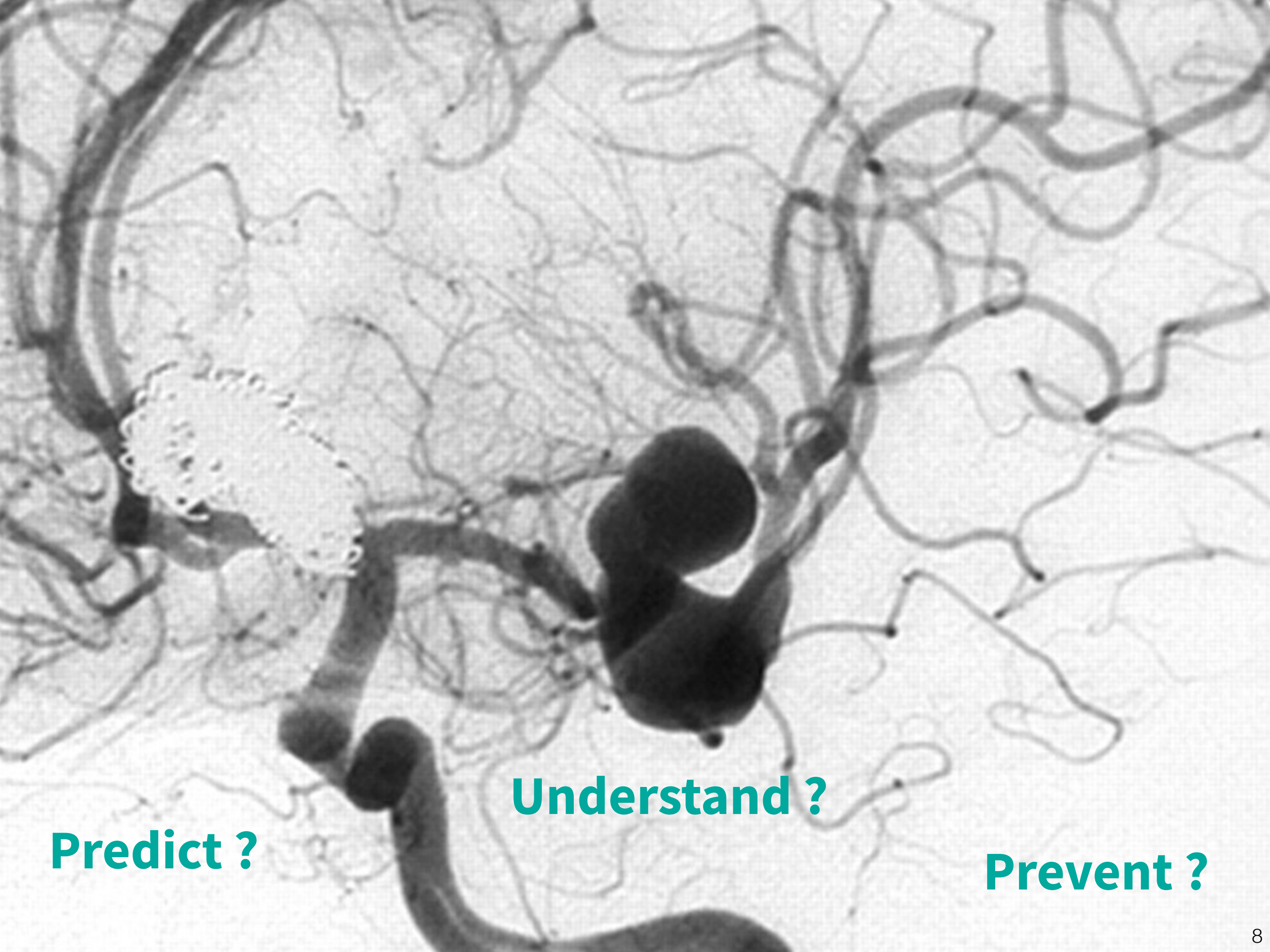
How are they treated ?

Regular MRI



+ endovascular surgery





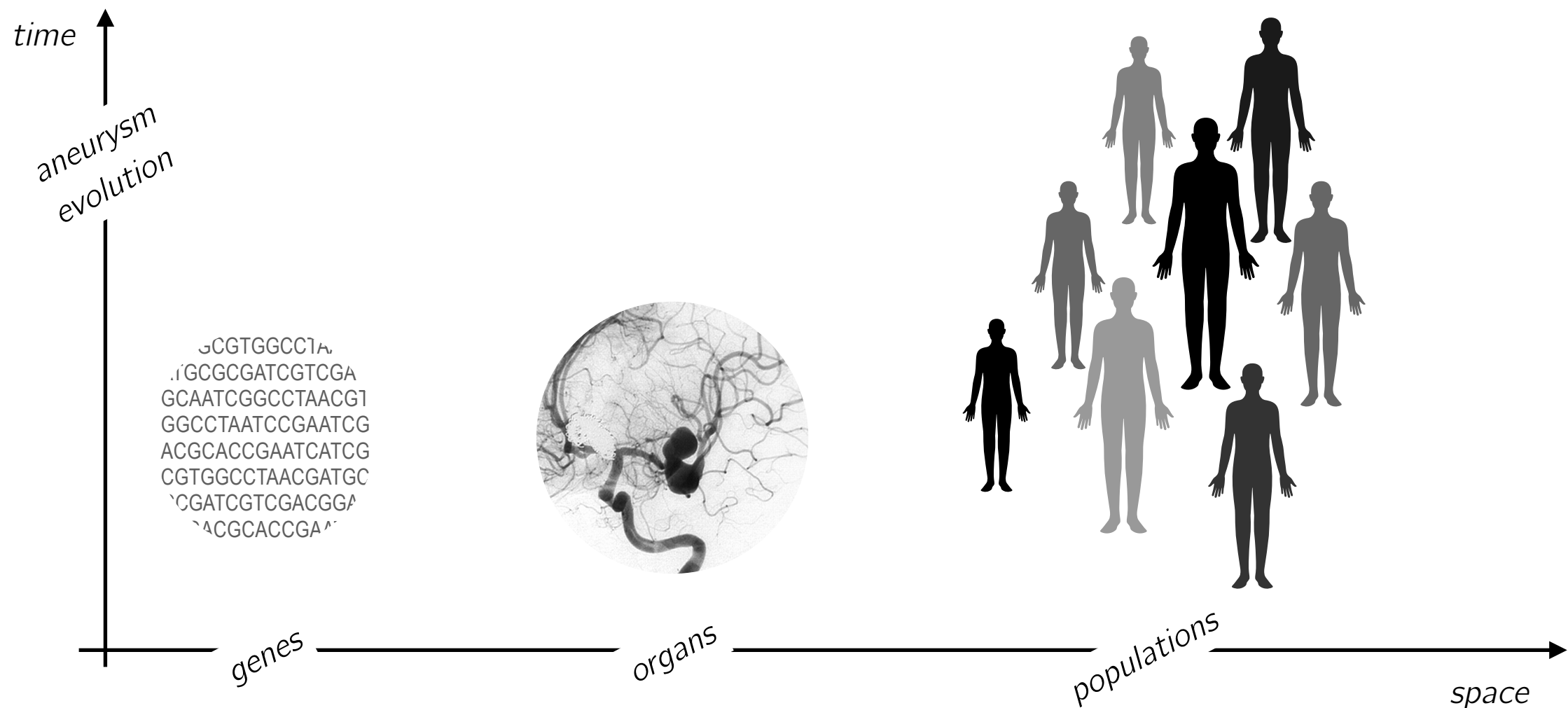
Understand ?

Predict ?

Prevent ?

Multi-factorial disease → multi-scale data

- ▶ Intracranial aneurysms: a complex & multifactorial disease
- ▶ Inter-disciplinary efforts needed for a better understanding of the pathology
- ▶ Specific data produced at very specific scales



Multi-centric study: 29 contributing centers

ICAN, UCAN, weCAN projects
+ recently funded eCAN RHU project



2023
4 000 ICA cases
TOF MRIs
800+ whole genomes

2024 +
> 3 000 patients per year



Using deep learning for an automatic detection and classification of the vascular bifurcations along the Circle of Willis

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Deep learning

ABSTRACT

Most of the intracranial aneurysms (ICA) occur on a specific portion of the cerebral vascular tree named the Circle of Willis (CoW). More particularly, they mainly arise onto fifteen of the major arterial bifurcations constituting this circular structure. Hence, for an efficient and timely diagnosis it is critical to develop some methods being able to accurately recognize each Bifurcation of Interest (BoI). Indeed, an automatic extraction of the bifurcations presenting the higher risk of developing an ICA would offer the neuroradiologists a quick glance at the most alarming areas. Due to the recent efforts on Artificial Intelligence, Deep Learning turned out to be the best performing technology for many pattern recognition tasks. Moreover, various methods have been particularly designed for medical image analysis purposes. This study intends to assist the neuroradiologists to promptly locate any bifurcation presenting a high risk of ICA occurrence. It can be seen as a Computer Aided Diagnosis scheme, where the Artificial Intelligence facilitates the access to the regions of interest within the MRI. In this work, we propose a method for a fully automatic detection and recognition of the bifurcations of interest forming the Circle of Willis. Several neural networks architectures have been tested, and we thoroughly evaluate the bifurcation recognition rate.

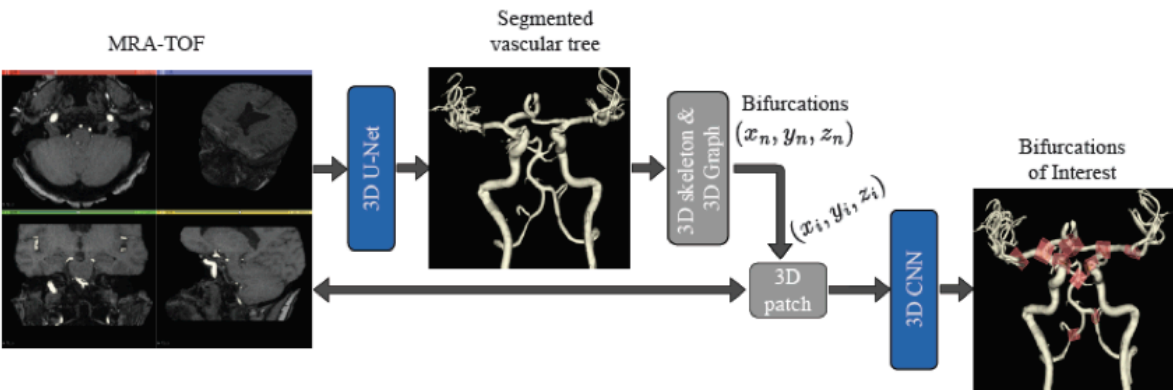
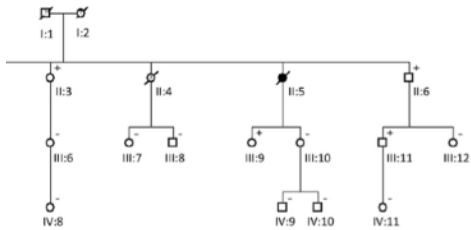


Fig. 1: General flowchart of the bifurcation recognition process.

Rare Coding Variants in *ANGPTL6* Are Associated with Familial Forms of Intracranial Aneurysm

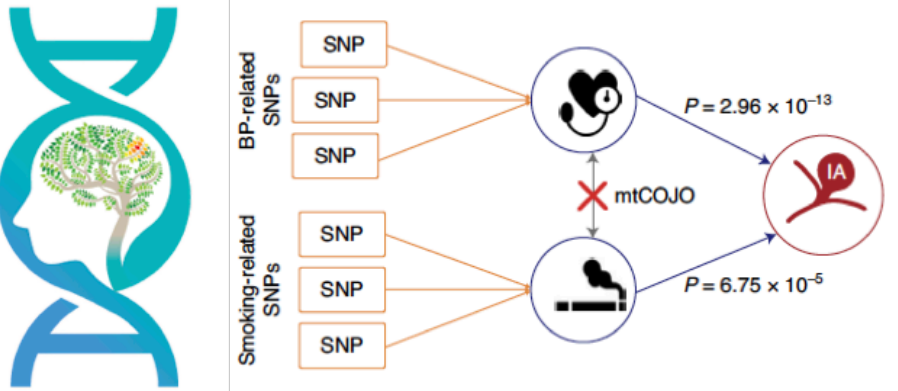
Romain Bourcier^{1,2} Solena Le Scouarnec¹ Stéphanie Bonnaud^{1,3} Matilde Karakachoff^{1,3} Emmanuelle Bourcereau³ Sandrine Heurtebise-Chrétien¹ Céline Menguy¹ Christian Dina^{1,3} Floriane Simonet^{1,3} Alexis Moles⁴ Cédric Lenoble² Pierre Lindenbaum¹ Stéphanie Chatel^{1,3} Bertrand Isidor⁵ Emmanuelle Génin⁶ Jean-François Deleuze⁷ Jean-Jacques Schott^{1,3} Hervé Le Marec^{1,3} ICAN Study Group, Gervaise Loirand^{1,3,8} Hubert Desal^{1,2,8,*} and Richard Redon^{1,3,8,*}

Intracranial aneurysms (IAs) are acquired cerebrovascular abnormalities characterized by localized dilation and wall thinning in intracranial arteries, possibly leading to subarachnoid hemorrhage and severe outcome in case of rupture. Here, we identified one rare nonsense variant (c.1378A>T) in the last exon of *ANGPTL6* (Angiopoietin-Like 6)—which encodes a circulating pro-angiogenic factor mainly secreted from the liver—shared by the four tested affected members of a large pedigree with multiple IA-affected case subjects. We showed a 50% reduction of *ANGPTL6* serum concentration in individuals heterozygous for the c.1378A>T allele (p.Lys460Ter) compared to relatives homozygous for the normal allele, probably due to the non-secretion of the truncated protein produced by the c.1378A>T transcript. Sequencing *ANGPTL6* in a series of 94 additional index case subjects with familial IA identified three other



Filtering steps	Remaining variants
Individuals :	III-1 III-5
Functional variants	25,674 23,456
MAF < 0.1% in ExAC (NFE)	456 436
Shared by III.1 & III.5	29
In IBD haplotypes	10
Shared by all affected	8
Predicted LOF	1 (<i>ANGPTL6</i>)

Genome-wide association study of intracranial aneurysms identifies 17 risk loci and genetic overlap with clinical risk factors

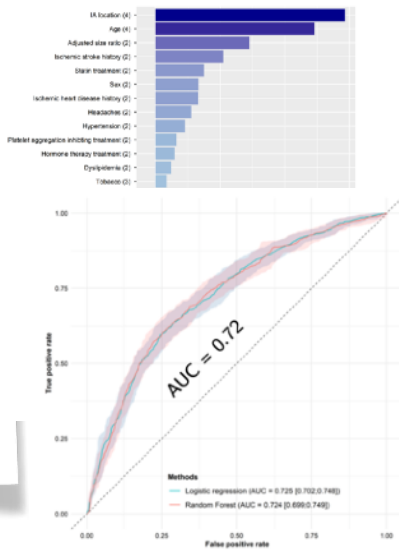


Observational Study > J Neurol Neurosurg Psychiatry. 2021 Feb;92(2):122-128.
doi: 10.1136/jnnp-2020-324371. Epub 2020 Oct 23.

Location of intracranial aneurysms is the main factor associated with rupture in the ICAN population

Olivia Rousseau¹, Matilde Karakachoff¹, Alban Gaignard², Lise Bellanger³, Philippe Bijlenga⁴, Pacôme Constant Diti Beaupils¹, Vincent L'Allinec⁵, Olivier Levrier⁶, Pierre Aguetaz⁷, Jean-Philippe Desilles⁸, Caterina Michelozzi⁹, Gaultier Marnat¹⁰, Anne-Clémence Vion², Gervaise Loirand², Hubert Desal¹¹, Richard Redon², Pierre-Antoine Gourraud¹, Romain Bourcier¹², ICAN Investigators

Collaborators, Affiliations + expand
PMID: 33097563 DOI: 10.1136/jnnp-2020-324371



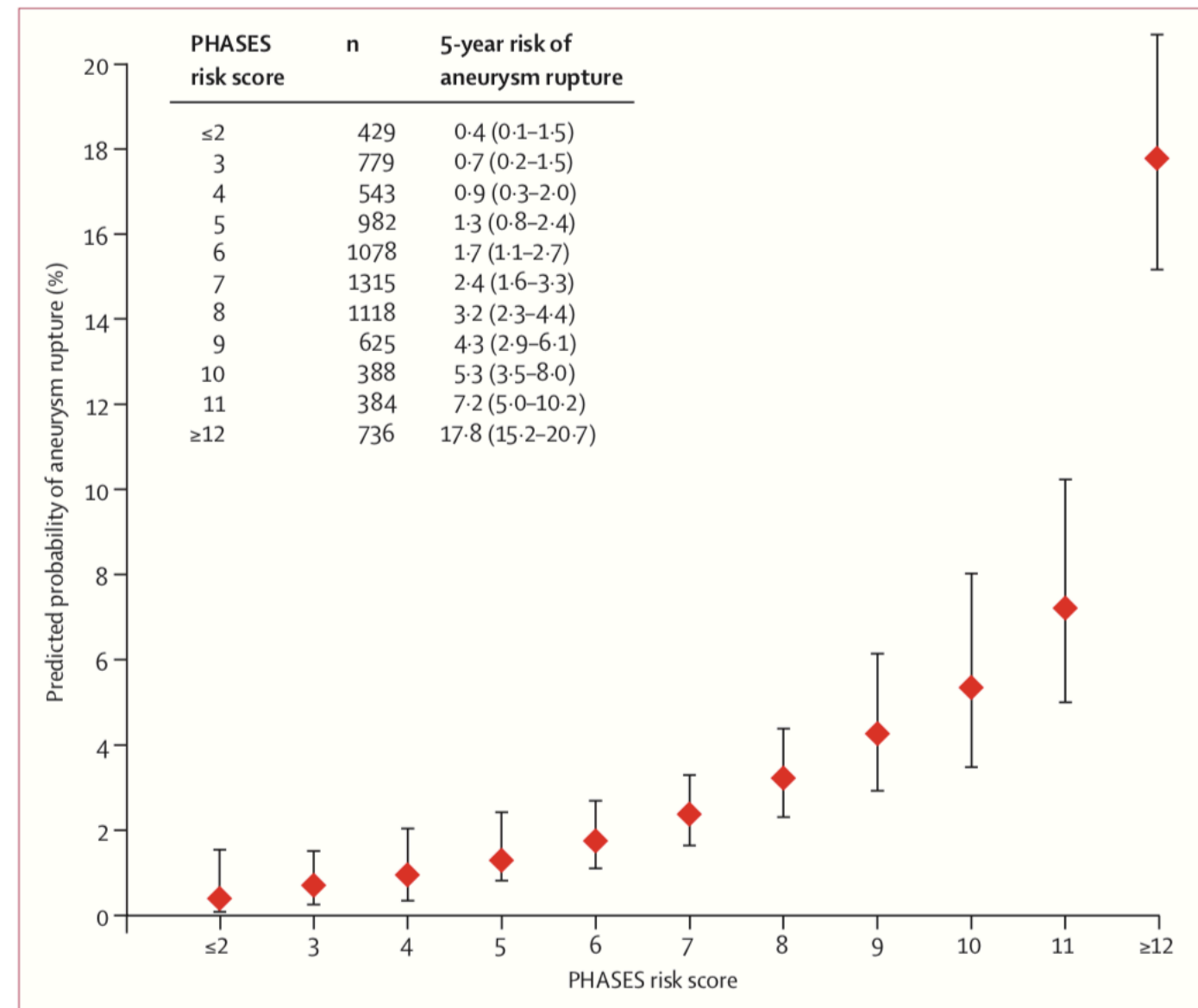


Olivia Rousseau

Is it possible to **predict** IA
rupture from **clinical data** ?

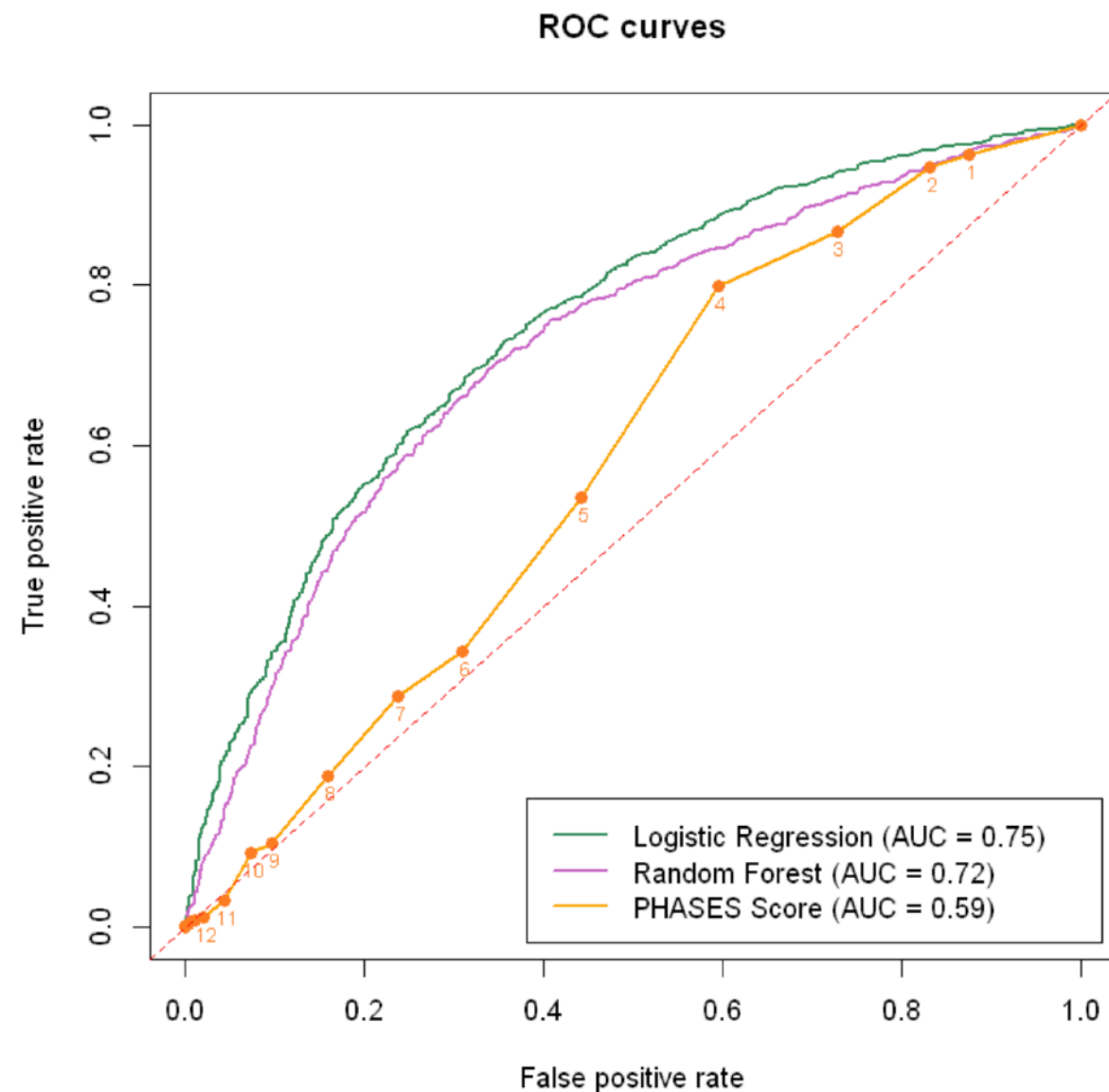
PHASES score

PHASES aneurysm risk score	Points
(P) Population	
North American, European (other than Finnish)	0
Japanese	3
Finnish	5
(H) Hypertension	
No	0
Yes	1
(A) Age	
<70 years	0
≥70 years	1
(S) Size of aneurysm	
<7.0 mm	0
7.0–9.9 mm	3
10.0–19.9 mm	6
≥20 mm	10
(E) Earlier SAH from another aneurysm	
No	0
Yes	1
(S) Site of aneurysm	
ICA	0
MCA	2
ACA/Pcom/posterior	4



Greving et. al. 2014 : Development of the PHASES score for prediction of risk of rupture of intracranial aneurysms: a pooled analysis of six prospective cohort studies.

Applying PHASES score to ICAN clinical data



- Unfortunately, poor prediction accuracy for PHASES (0.59)
- Better prediction accuracy with state-of-the-art Machine Learning / Statistical methods + other variables
- **Predictions are still failing in around 25% of cases ...**

integrating imaging or genomic markers ?

Intracranial aneurysms, current challenges

Complex and **multi-scale** biological phenomena involved :

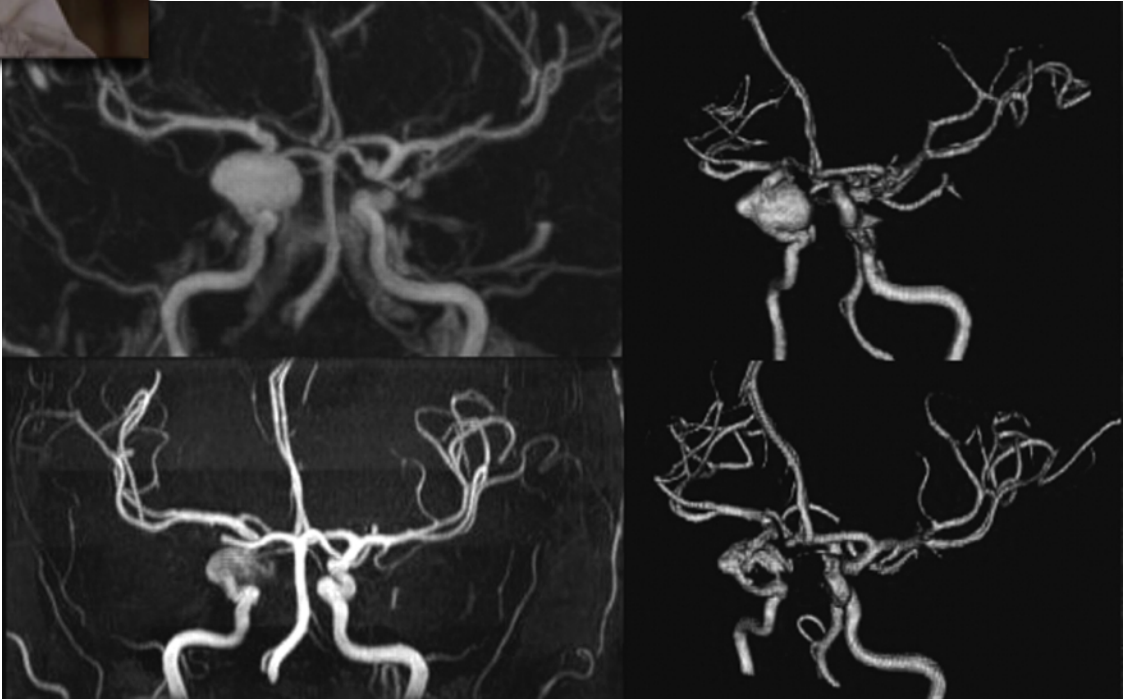
- Impact of life habits ? Image-based evaluation ?
- Any genetic variant specific to image-based phenotypes ?
- Mirror aneurysms patient sub-grouping ? (Vincent Lallinec)

Needs for **jointly** exploiting genomic data, MR imaging and clinical observations ...

... to better **understand** and **predict** evolution schemes.

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	39458_2001	39458_2003	39458_2005			
39458_2007		39458_2009		39458_2012		39458_3003		39458_39461	39458_39462	39458_39465	39458_39466			
39458_CD12099		39458_CD12100		39458_CD12252		39458_CD12350		39458_CD12380	39458_CD12406	39458_CD12466	39458_CD12480			
39458_CD12501		39458_CD12586		39458_CD12587		39458_CD12588		39458_CD12704	39458_CD13016	39458_CD13018	39458_CD13178			
39458_CD13363														
1	752566	rs3094315		A	G	.	.	PR	GT	./.	./.	./.	./.	./.
./.	./.	./.	./.	./.	0/1	0/0	0/1	0/1	0/1	0/0	0/1	0/1	0/1	0/1
0/1	0/0	0/0	0/1	0/1	1/1									
1	754105	rs12184325		C	.	.	.	PR	GT	./.	./.	./.	./.	./.
./.	./.	./.	./.	./.	0/0	0/0	0/0	0/0	0/0	0/0	0/0	0/0	0/0	0/0
0/0	0/0	0/0	0/0	0/0	0/0									
1	754182	rs3131969		G	A	.	.	PR	GT	./.	./.	./.	./.	./.
./.	./.	./.	./.	./.	0/1	0/0	0/1	0/1	0/1	0/0	0/1	0/1	0/1	0/1
0/1	0/0	0/0	0/1	0/1	1/1									
1	758626	rs3131954		T	.	.	.	PR	GT	./.	./.	./.	./.	./.
./.	./.	./.	./.	./.	0/0	0/0	0/0	0/0	0/0	0/0	0/0	0/0	0/0	0/0
0/0	0/0	0/0	0/0	0/0	0/0									
1	768448	rs12562034		G	A	.	.	PR	GT	./.	./.	./.	./.	./.
./.	./.	./.	./.	./.	0/0	0/0	0/0	0/1	0/0	0/1	0/1	0/0	0/0	0/1
0/0	0/0	0/0	0/0	0/0	0/0									
1	792534	rs2518996		G	.	.	.	PR	GT	./.	./.	./.	./.	./.
./.	./.	./.	./.	./.	0/0	0/0	0/0	0/0	0/0	0/0	0/0	0/0	0/0	0/0
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./.	./.	./.	./.	./.	0/1	0/0	0/0	0/1	0/0	1/1	0/1	0/0	0/0	0/1
0/1	0/0	0/0	0/0	0/0	0/0									
1	798959	rs11240777		G	A	.	.	PR	GT	./.	./.	./.	./.	./.
./.	./.	./.	./.	./.	0/1	0/0	0/0	0/1	0/1	1/1	0/1	0/1	0/1	0/1
0/1	0/0	0/0	0/0	0/1	0/0									

première atteinte	circonstances d'admission	consommation d'alcool	quantité d'alcool	statut phénotypique	nombre d'antécédents	date du 1er diagnostic	AIC 1 Grand axe	AIC 1 Localisation	AIC 2 Grand axe	AIC 2 Localisation
Fortuite	Sévérité > 3 ans	96.6	certain	1.0	1/23/16	7.0	5			
Déclatage familial	Tobacq sans add	277.2	certain	1.0	3/3/10	4.0	10			
Fortuite	Sévérité < 3 ans	106.2	certain	3.0	12/28/10	6.0	4	14.0	1	
Rupture AIC	Jamaïque fumé	0.0	certain	2.0	4/3/08	7.0	M.C.A. Gênes	5.0	M.C.A. Gênes	
Rupture AIC	Sévérité > 3 ans		certain	4.0	1/15/17		18		17	
Fortuite	Jamaïque fumé	25.2	certain	1.0	1/23/16	5.0	21			
Fortuite	Sévérité < 3 ans	80.4	certain	3.0	5/23/15	5.0	18	6.9	118	
Rupture AIC	Jamaïque fumé		certain	1.0	5/1/05	6.0	118			
Rupture AIC	Jamaïque fumé		certain	1.0	1/16/10	9.0	107			
Fortuite	Tobacq sans add		certain	1.0		7.0	105			
Rupture AIC	Sévérité > 3 ans	56.8	certain	1.0	1/5/10	2.0	10			
Rupture AIC	Jamaïque fumé		certain	1.0		4.0	10			
Fortuite	Jamaïque fumé	46.2	certain	1.0		8.0	10			
Fortuite	Sévérité > 3 ans		certain	2.0		8.0	18	5.0	10	
Fortuite	Jamaïque fumé		certain	1.0	8/13/16	28.0	104			
Déclatage familial	Jamaïque fumé		certain	1.0	1/1/08	8.0	118			
Fortuite	Tobacq sans add		certain	1.0		8.0	20			
Rupture AIC	Tobacq sans add	25.2	certain	4.0	3/3/18	7.0	17	4.0	104	18.0
Rupture AIC	Tobacq sans add	336.0	certain	1.0		5.0	132			105.0
Rupture AIC	Jamaïque fumé		certain	3.0	1/1/10	3.0	10	4.0	10	3.0
Rupture AIC	Tobacq sans add		certain	2.0	5/23/05	12.0	10	3.0	105	10.0
Déclatage familial	Sévérité < 3 ans	376.0	certain	1.0	1/1/10	2.0	10			
Rupture AIC	Jamaïque fumé		certain	1.0	2/5/12	20.0	6			
Rupture AIC	Jamaïque fumé		certain	1.0	1/1/03	6.0	6			
Rupture AIC	Tobacq sans add	84.0	certain	3.0	1/3/16	4.0	10	2.0	118	2.0
Rupture AIC	Jamaïque fumé		certain	2.0	3/15/16	5.0	118	1.5	18	108.0
Déclatage familial	Sévérité > 3 ans		certain	1.0		2.0	18			
Rupture AIC	Tobacq sans add	0.0	certain	3.0	6/14/18	2.0	18	5.0	21	15.0
Rupture AIC	Sévérité > 3 ans	12.6	certain	3.0	11/1/14	6.0	10	2.0	18	3.0
Rupture AIC	Sévérité > 3 ans	25.2	certain	1.0	8/2/16	2.0	10			119.0
Fortuite	Jamaïque fumé		certain	1.0	1/23/16					
Rupture AIC	Sévérité < 3 ans	470.4	certain	2.0	3/24/10	2.0	10	3.0	12	
Fortuite	Tobacq sans add	12.6	certain	1.0	1/1/04	8.0	10			
Fortuite	Tobacq sans add	0.0	certain	2.0	1/18/15	11.0	118	5.5	18	
Fortuite	Jamaïque fumé		certain	1.0						
Rupture AIC	Jamaïque fumé	0.0	certain	4.0	1/18/14	7.0	18	4.0	108	4.0
Rupture AIC	Jamaïque fumé	0.0	certain	1.0						118.0
Fortuite	Jamaïque fumé	0.0	certain	2.0		3.5	12	2.5	18	
Rupture AIC	Jamaïque fumé	0.0	certain	1.0	7/23/12	5.0	132			
Fortuite	Tobacq sans add	42.0	certain	2.0	4/24/15	5.5	105	13.0	5	
Déclatage familial	Sévérité < 3 ans	21.0	certain	1.0	1/1/10	7.0	105			
Fortuite	Tobacq sans add	11.76	certain	2.0	6/23/16	8.0	109			
Rupture AIC	Sévérité < 3 ans	35.28	certain	1.0	6/30/10	2.0	5	3.0	185	
Fortuite	Jamaïque fumé	11.76	certain	1.0	8/4/18	5.0	111			
Rupture AIC	Sévérité > 3 ans	0.0	certain	1.0	4/1/18	8.5	121			
Rupture AIC	Jamaïque fumé	11.76	certain	1.0	2/13/14	8.5	10			
Rupture AIC	Sévérité > 3 ans	35.28	certain	1.0	5/5/15	6.7	10			
Fortuite	Jamaïque fumé	11.76	certain	1.0	2/23/16	6.0	111			
Déclatage familial	Tobacq sans add	11.76	certain	1.0	4/1/10	9.0	107			
Fortuite	Sévérité > 3 ans	202.20	certain	2.0	1/1/14	4.4	17	5.2	18	



How to jointly analyse
these diverse datasets ?

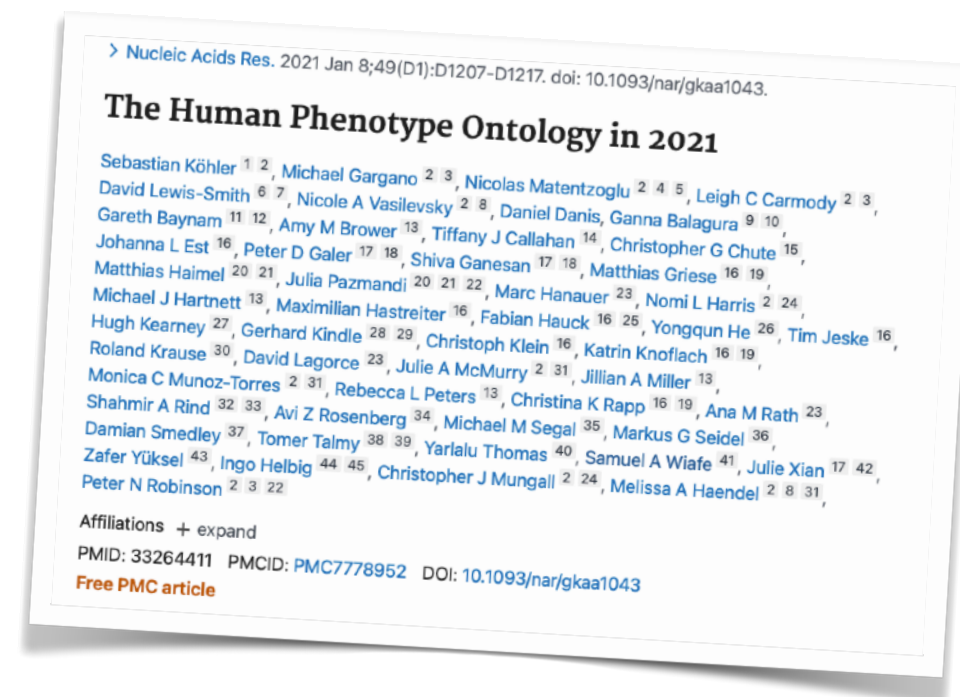
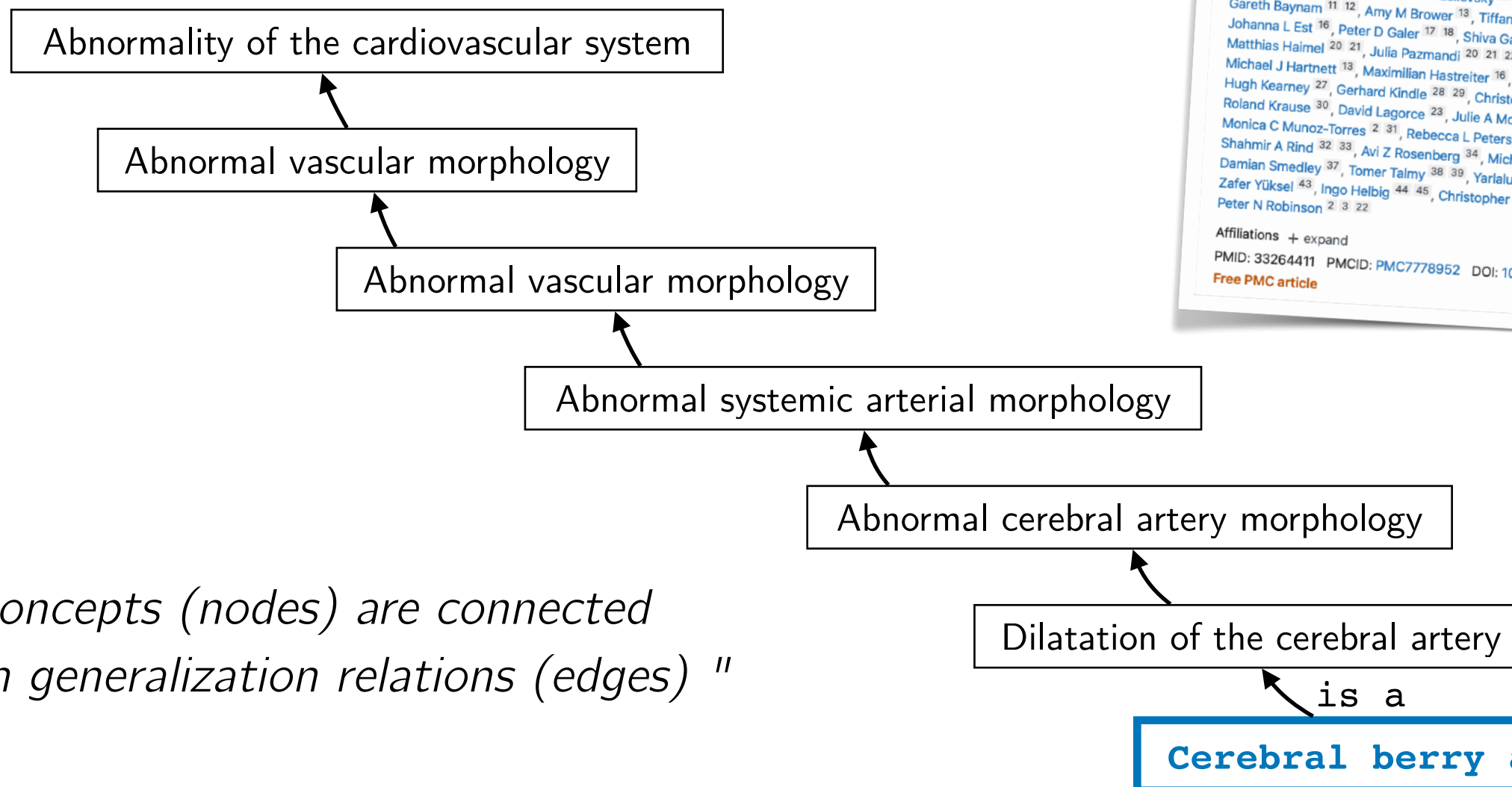
① what is observed with data ?

Computational ontology

« a **formal specification** of a **shared conceptualization** » (Borst, 1997)

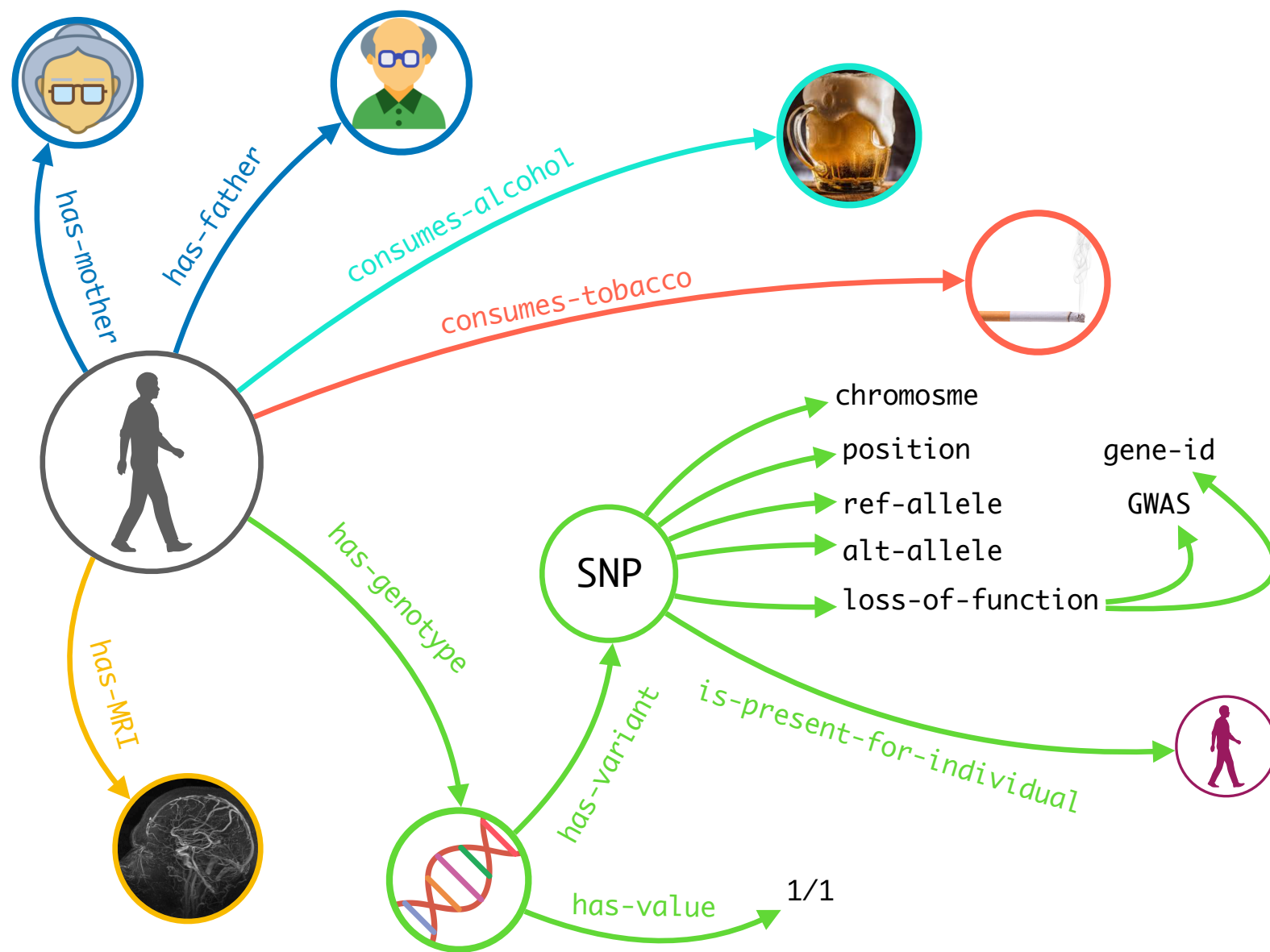
→ 1,049 life science ontologies registered in BioPortal (2023)

Human Phenotype Ontology



② how to **link** data from multiple scales ?

Knowledge graphs: « a collection of interlinked descriptions of things (real-world objects, abstract concepts, events, etc.) »



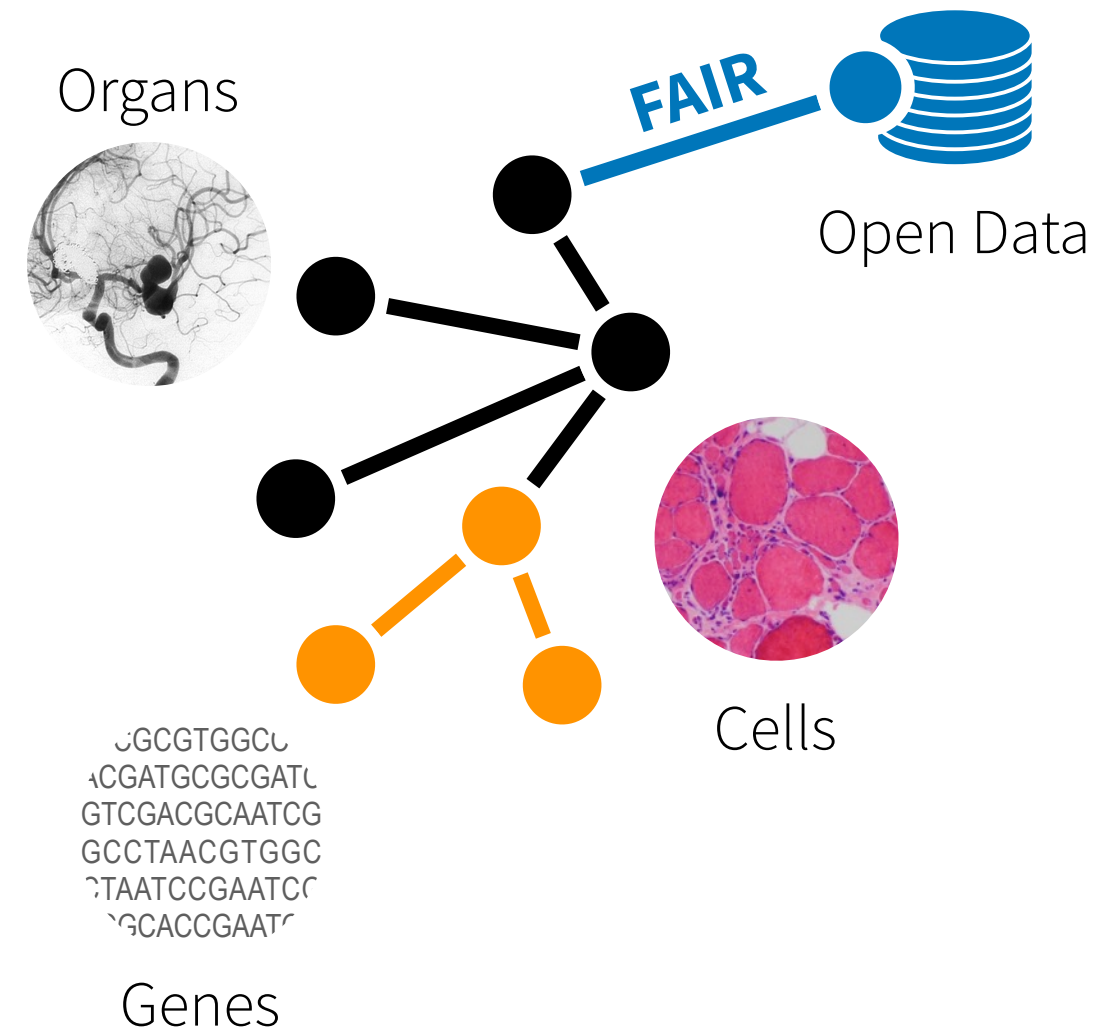
= **database:** information storage / extraction

= **graph:** network analysis

= **knowledge base** with formal semantics :
+ logical facts
+ logical inferences

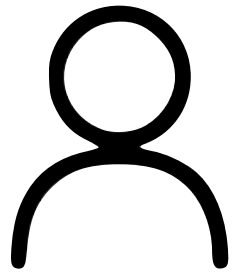
Bridging imaging-omics- clinical data

INEX-MED

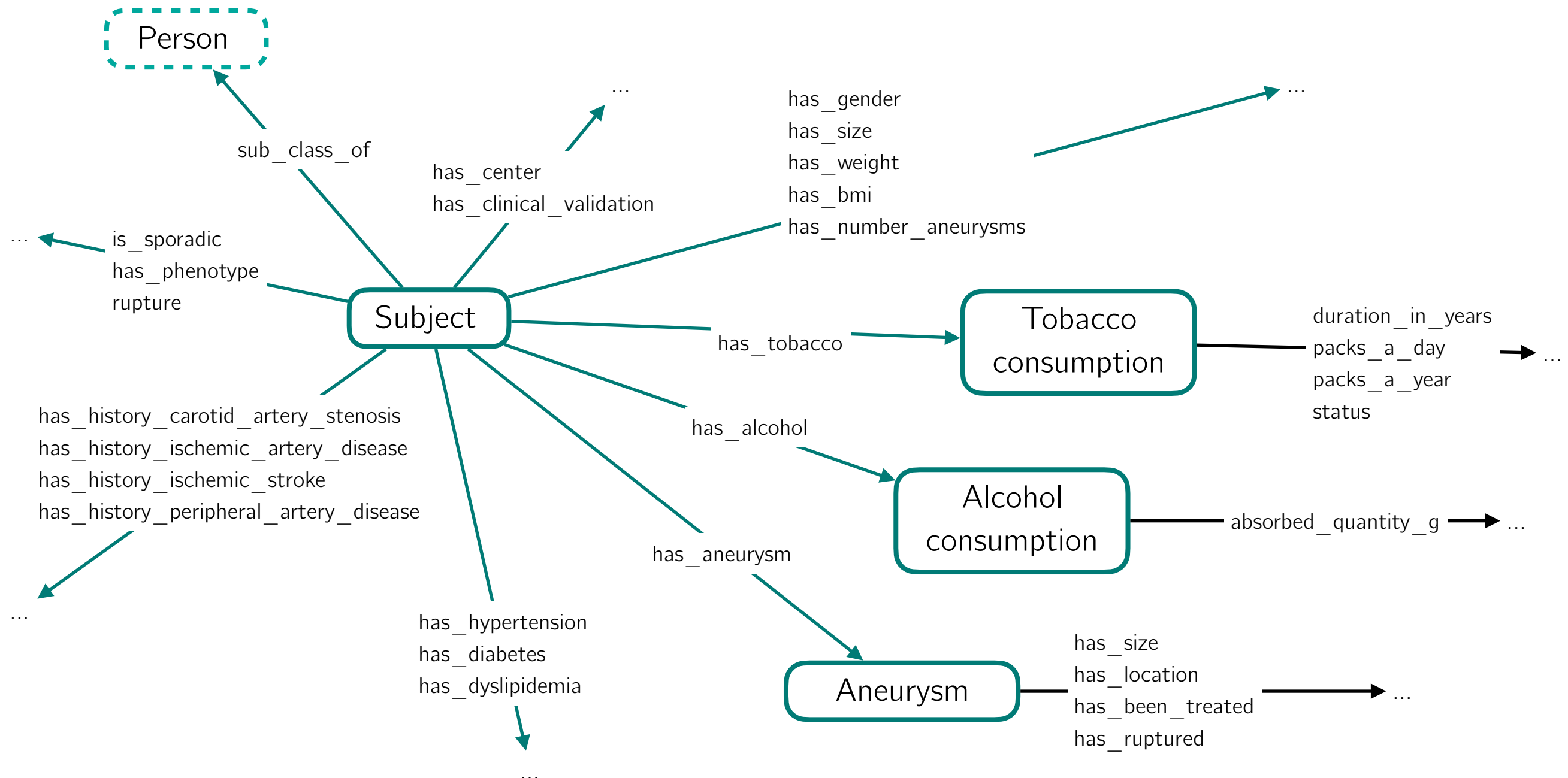


Multi-scale « Knowledge Graph »
145 Million facts

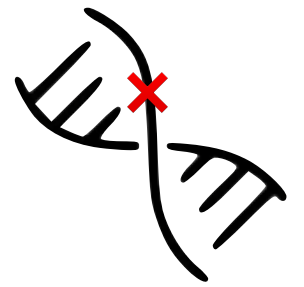
Clinical observations



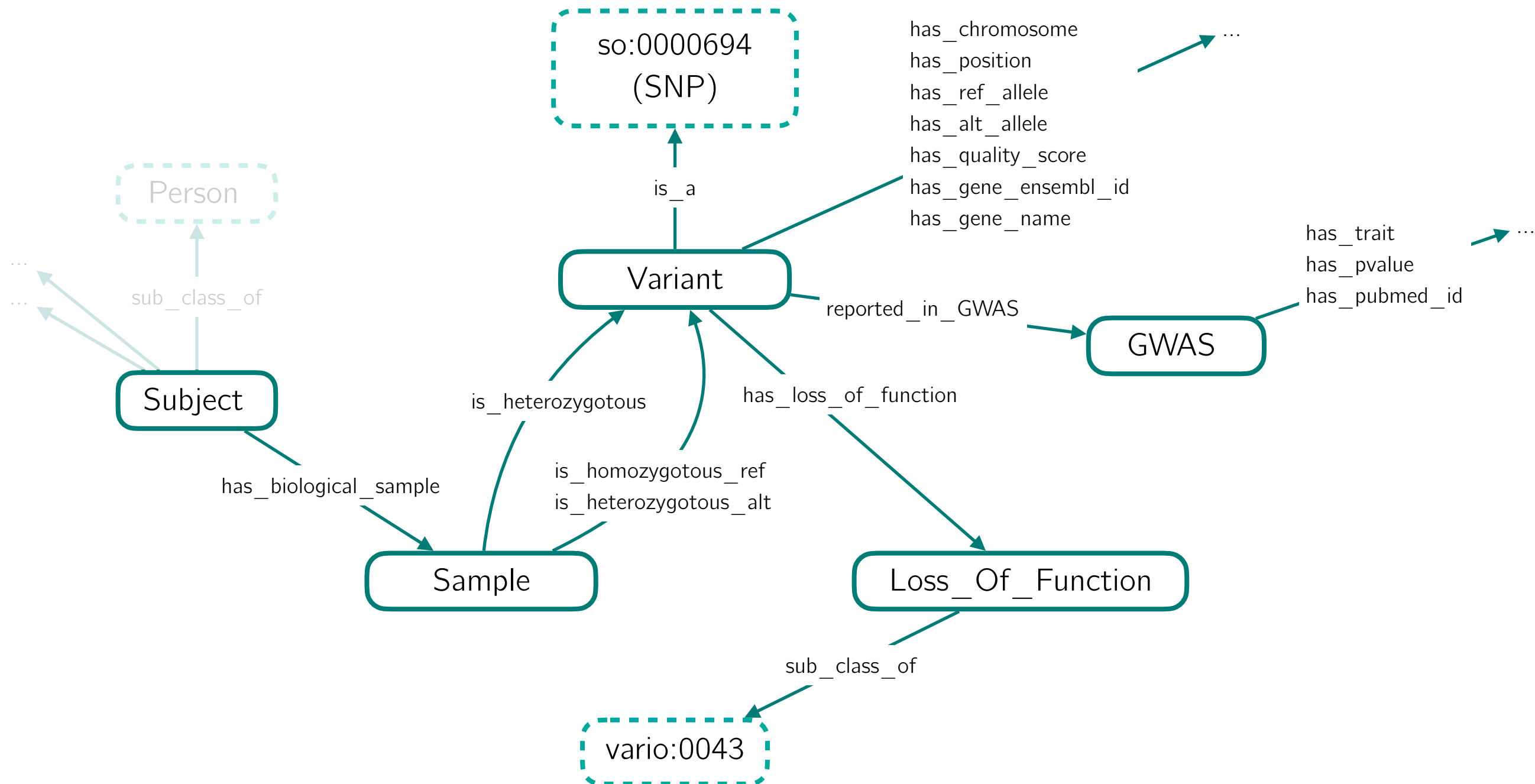
- Aim : representing key characteristics of ICA patients



Genomic markers



- Aim : representing associated genetic variations by **extending reference ontologies** (Sequence Ontology, VARIO)



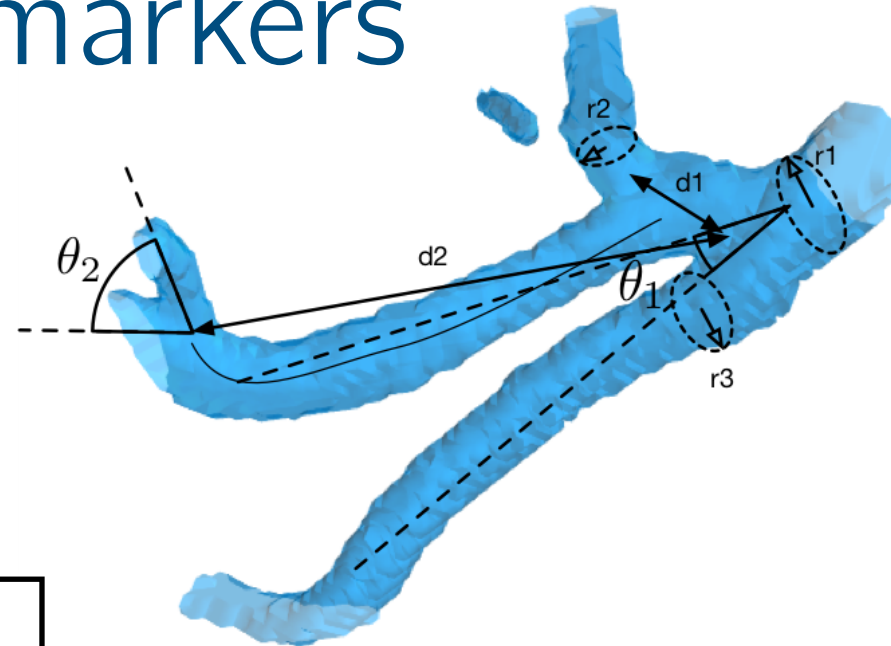
Imaging markers



Anass Nouri



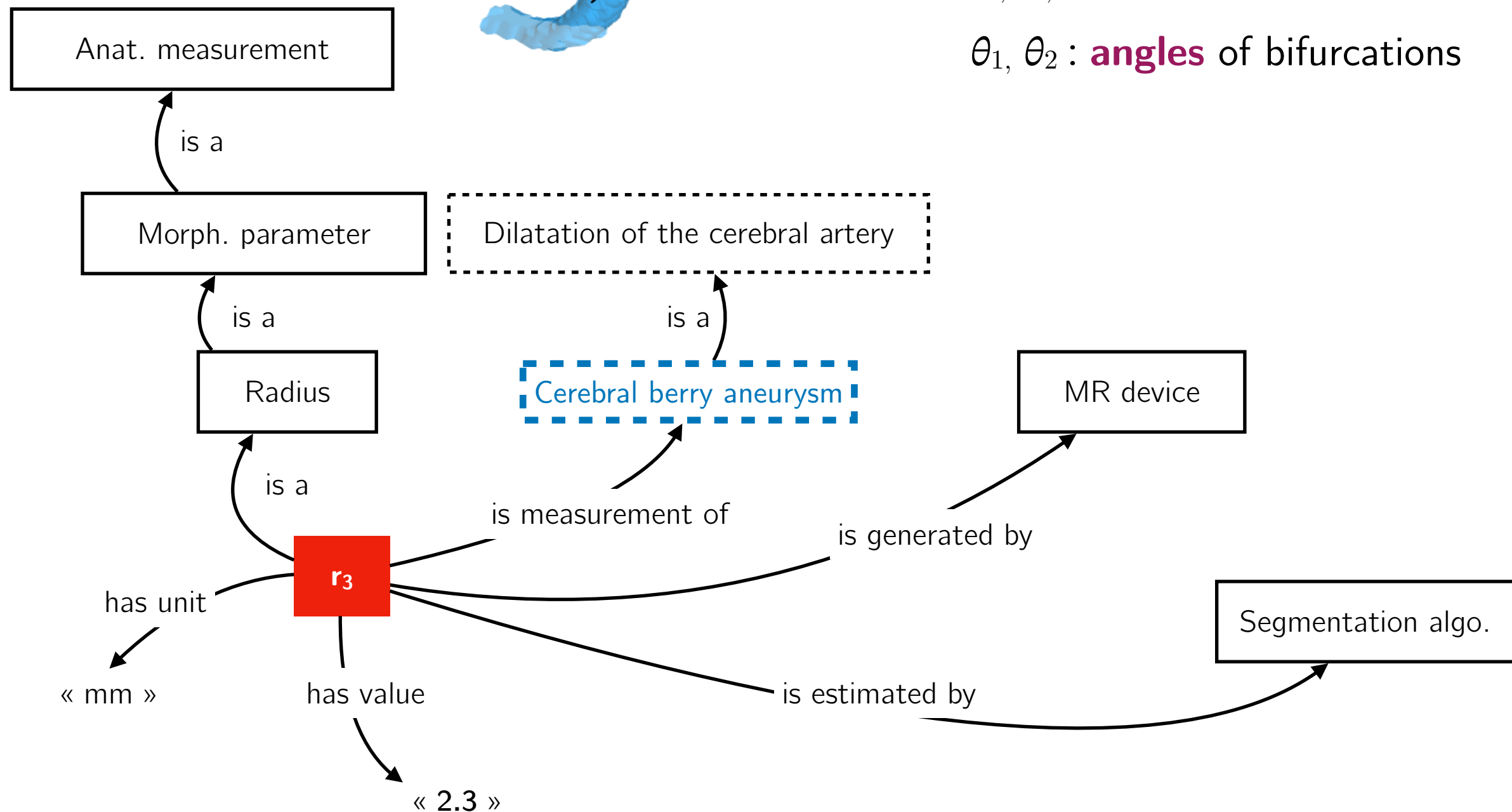
Florent Autrusseau



d_1, d_2 : **distances** between **bifurcations**

r_1, r_2, r_3 : **radius** of vessels

θ_1, θ_2 : **angles** of bifurcations



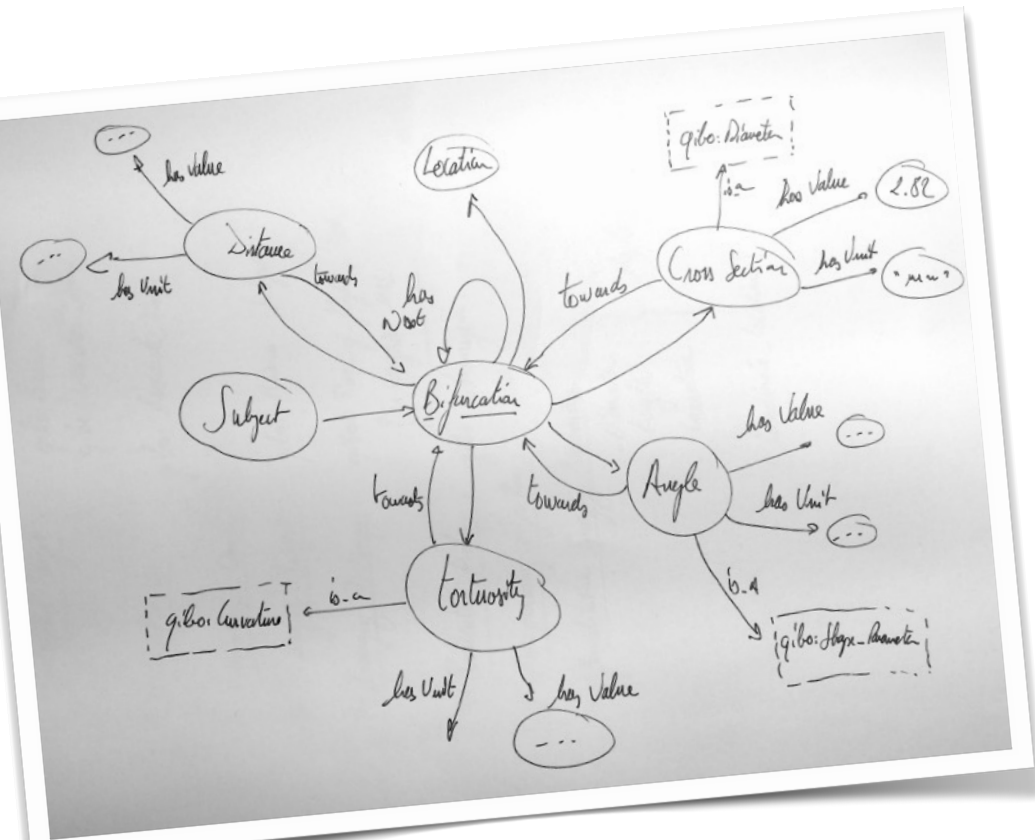
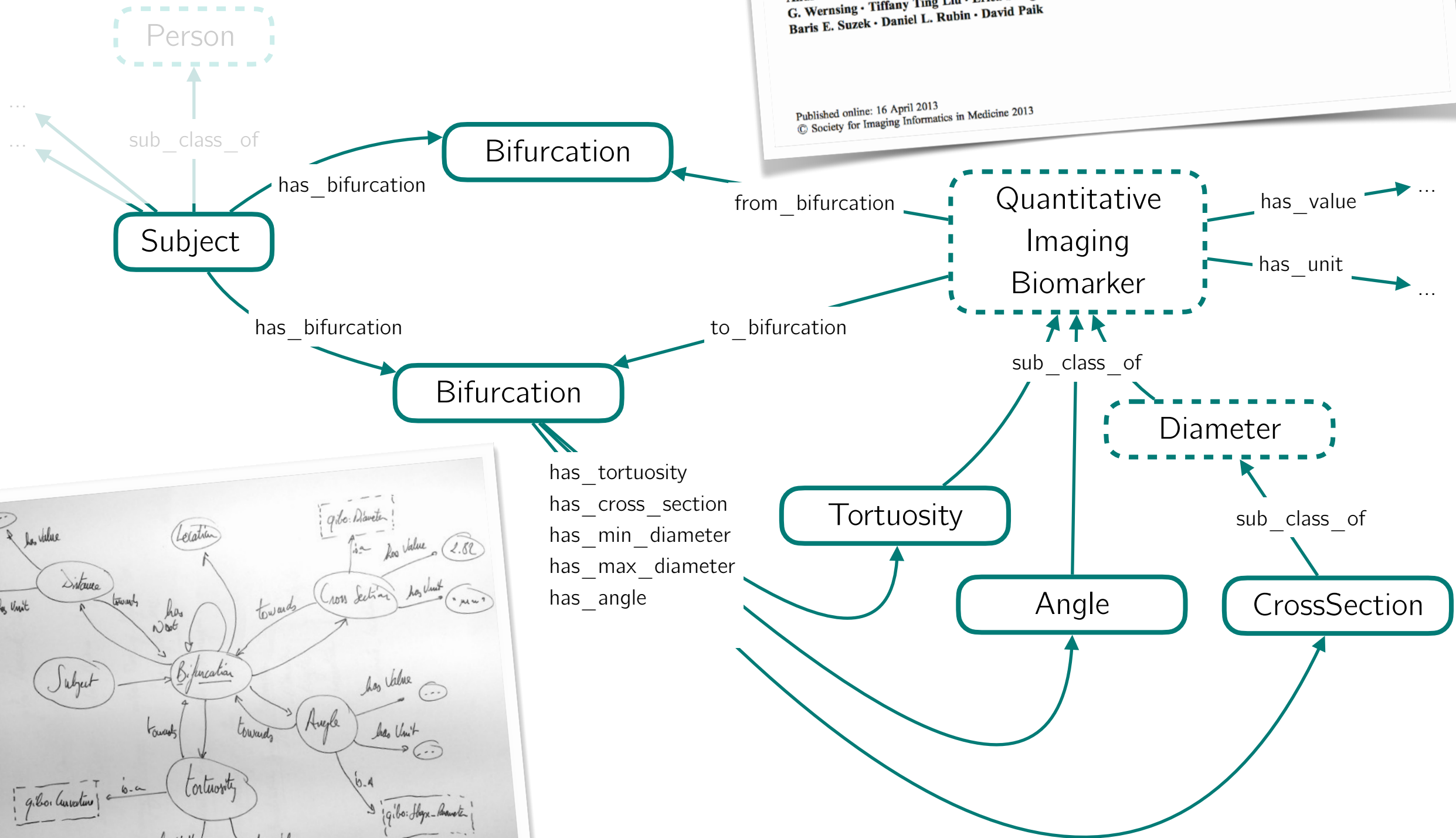
Imaging markers

J Digit Imaging (2013) 26:630–641
DOI 10.1007/s10278-013-9599-2

Quantitative Imaging Biomarker Ontology (QIBO) for Knowledge Representation of Biomedical Imaging Biomarkers

Andrew J. Buckler • M. Ouellette • J. Danagoulain •
G. Wernsing • Tiffany Ting Liu • Erica Savig •
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Searching
multi-scale data ?

Queries with graph patterns



"Give all patients with a mutation in ANGPTL6, in the case of a rupture, give me also the age and the location of the ruptured aneurysm."

```
SELECT * WHERE {  
  # Search variants  
  ?subject ican:has_dna_sample ?sample .  
  ?sample ican:is_heterozygous ?variant .  
  ?variant ican:has_chromosome "19" ;  
           ican:has_position ?position ;  
           ican:has_alt_allele ?alt_allele ;  
           ican:has_ref_allele ?ref_allele .  
  FILTER (?position > 10092337 && ?position < 10106407)  
  # dbSNP identifier  
  OPTIONAL { ?variant ican:has_dbSNP_id ?dbSNPid . }  
  # First aneurysm rupture: age and localisation  
  OPTIONAL { ?subject ican:age_first_rupture ?ageRupt . }  
  OPTIONAL { ?subject ican:loc_first_rupture ?locRupt . }  
}
```

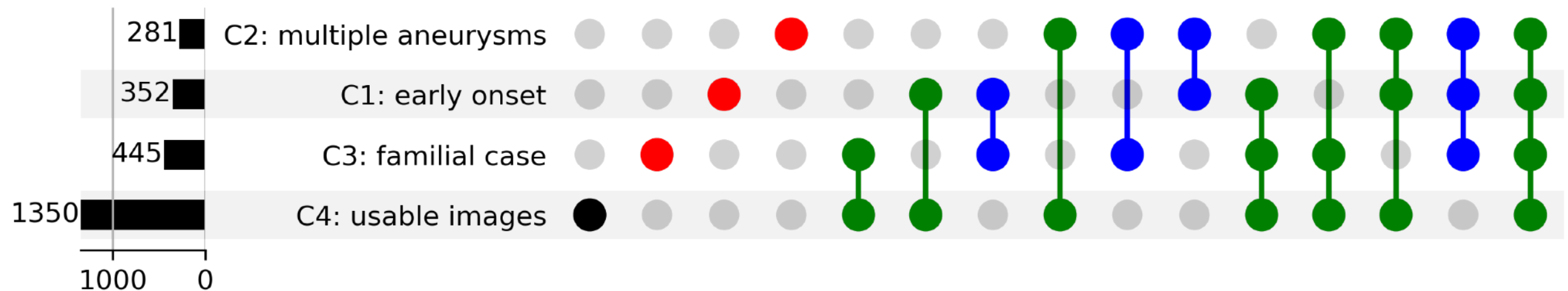
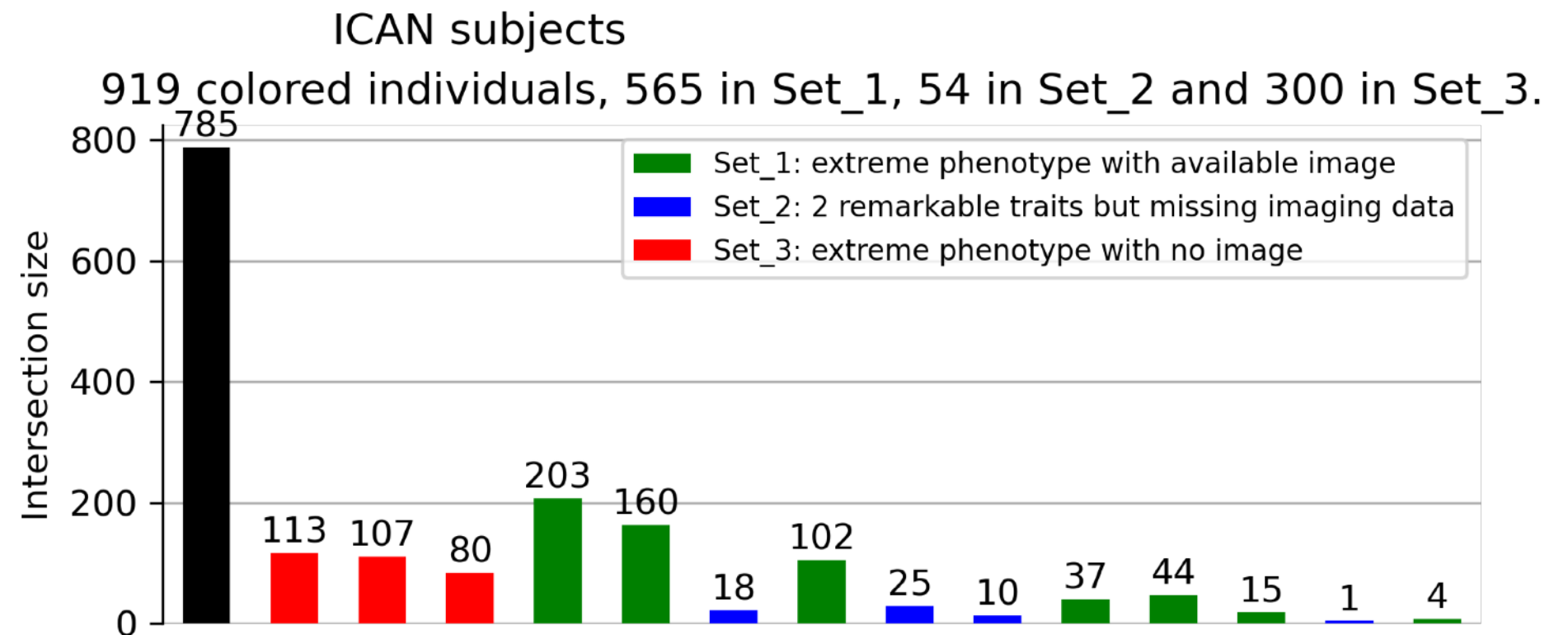
Select "extreme phenotypes" for genomic studies

usable images ?

< 30 years old ?

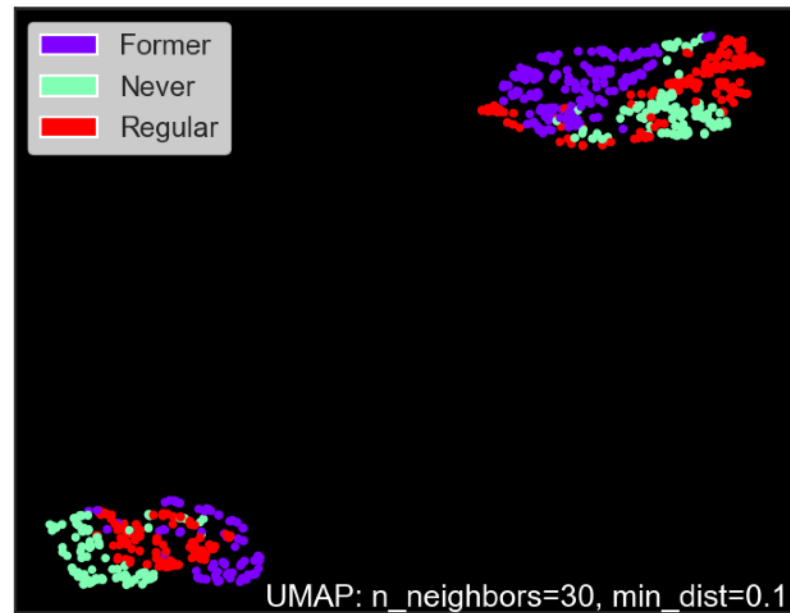
2+ aneurysms ?

familial forms ?

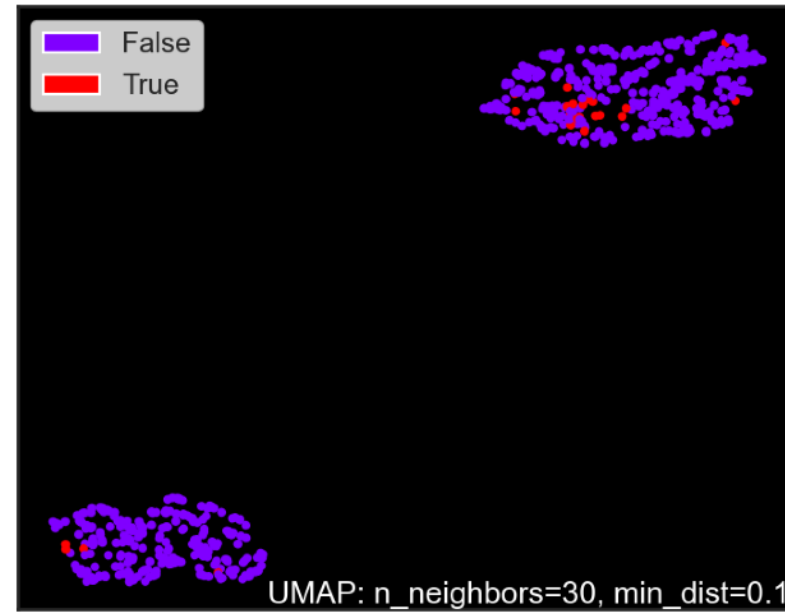


Train machine-learning models ...

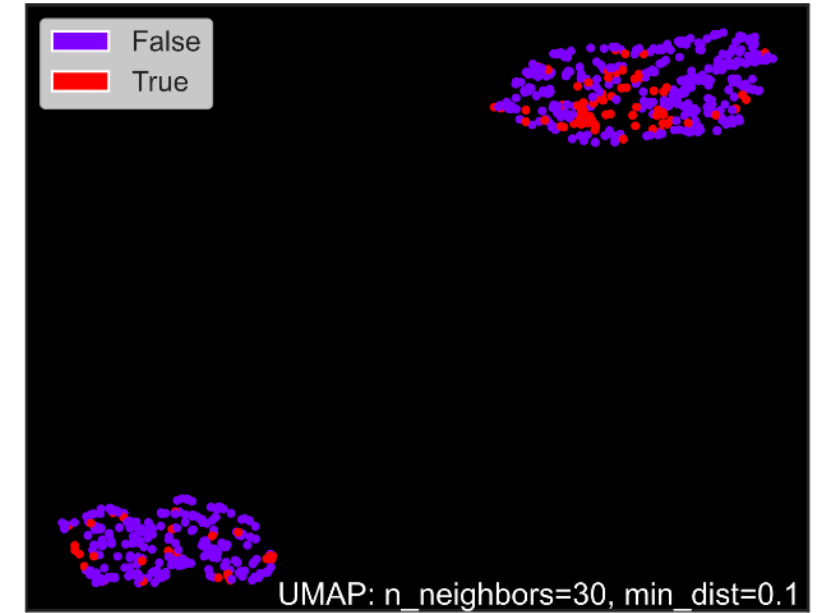
Tobacco consumption



Ischemic heart disease history



Platelet aggregation inhibiting treatment



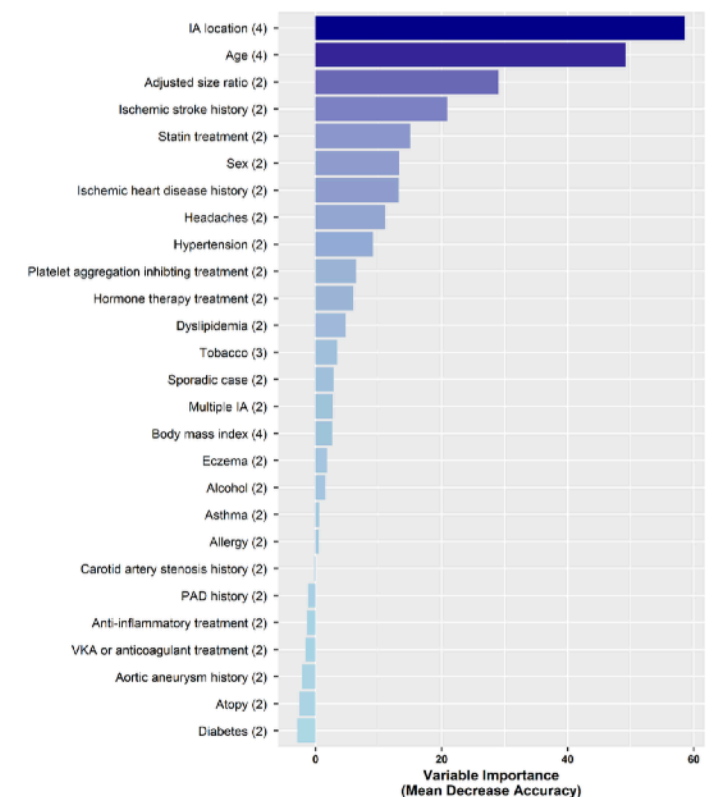
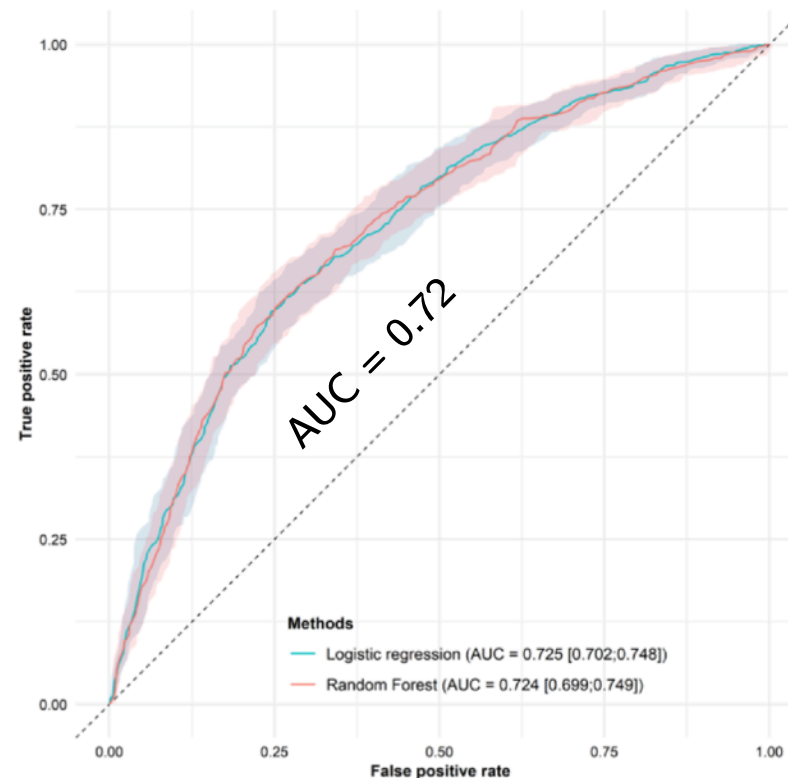
Observational Study > J Neurol Neurosurg Psychiatry. 2021 Feb;92(2):122-128.
doi: 10.1136/jnnp-2020-324371. Epub 2020 Oct 23.

Location of intracranial aneurysms is the main factor associated with rupture in the ICAN population

Olivia Rousseau¹, Matilde Karakachoff¹, Alban Gaignard², Lise Bellanger³, Philippe Bijlenga⁴,
Pacôme Constant Dît Beaufils¹, Vincent L'Allinec⁵, Olivier Levrier⁶, Pierre Aguetaz⁷,
Jean-Philippe Desilles⁸, Caterina Michelozzi⁹, Gaultier Marnat¹⁰, Anne-Clémence Vion²,
Gervaise Loirand², Hubert Desal¹¹, Richard Redon², Pierre-Antoine Gourraud¹,
Romain Bourcier¹²; ICAN Investigators

Collaborators, Affiliations + expand

PMID: 33097563 DOI: 10.1136/jnnp-2020-324371



(controlled) access and sharing

INEX-MED data hub

GENERAL

Dashboard

Admin

SEARCH

Centers

Gender Distribution

Aneurysms Position

Genomic Variants

Query

AUTHENTICATION

Sign-in

Sign-out

INEX-MED data hub

Exploring clinical, genomics, and imaging data to better understand congenital myopathies and intracranial aneurysms

See the [official INEX-MED wiki page](#) for more information.

SUBJECTS	CLINICAL OBSERVATIONS	BIFURCATIONS	GENETIC VARIANTS
2,974	2,613	9,490	1,549,005

FACTS

143,745,265

Web demonstrator → <https://inexmed-api.univ-nantes.fr>

Related works

Unruptured IA: 51 "core" common data elements / 1254

Age value, Birth date, Birth sex assigned type, Body system category, Race USA category, Sex genotype type, Ethnicity USA category, Gender identity type, Gender type,

Aneurysm diameter maximum measurement, Aneurysm height measurement, Aneurysm morphology type, Aneurysm width measurement,

Hypertension indicator,

Tobacco current use indicator, Tobacco prior use indicator, Tobacco use started age value, Tobacco use stopped age value,

World Federation of Neurological Surgeons (WFNS) - grading system subarachnoid hemorrhage scale

Behavioral history assessment date and time,

Biospecimen collection other text, Biospecimen collection type,

Cerebrospinal fluid collection anatomic site, Cerebrospinal fluid collection anatomic site other text, Cerebrospinal fluid collection method other text, Cerebrospinal fluid collection method type,

Data collected date and time, Data source,

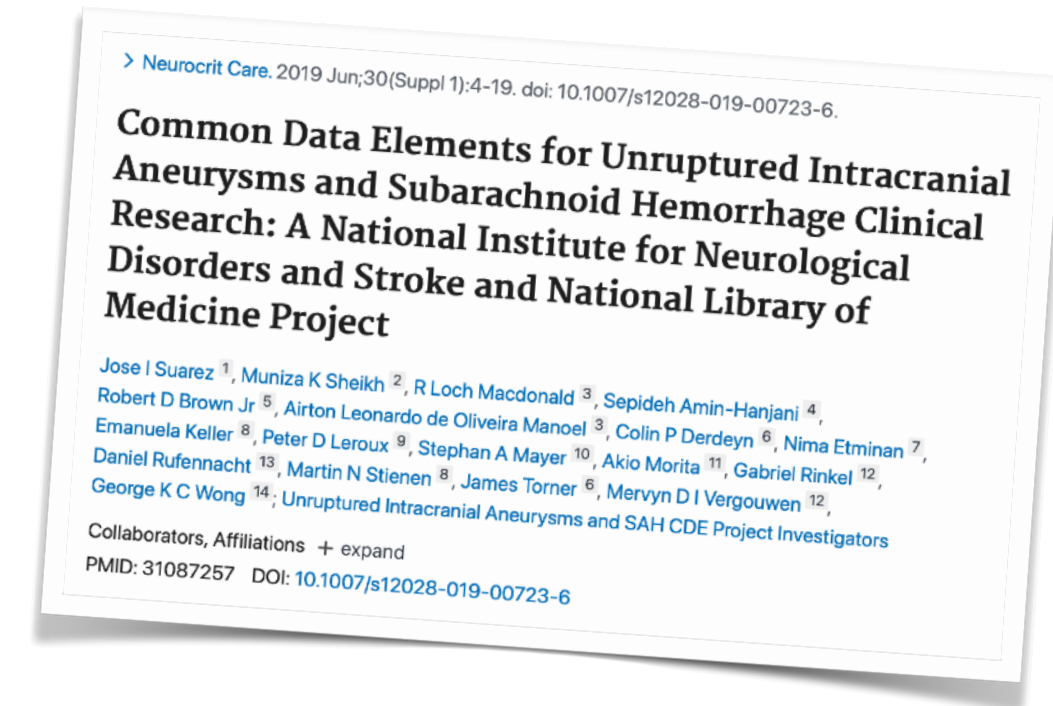
History data not obtained reason,

Imaging modality other text, Imaging modality type, Imaging modality vessel imaging angiography type, Imaging vessel angiography arterial anatomic site, Imaging vessel angiography arterial result,

Intervention endovascular status, Intervention surgical status, Lab specimen collection date and time,

Medical history condition SNOMED CT code, Medical history condition text, Medical history taken date and time,

Specimen acquisition anatomical site, Specimen baseline collection indicator, Specimen biologic source type, Specimen centrifugation duration, Specimen centrifugation revolutions per minute value, Specimen centrifugation temperature value, Specimen collection tube other text, Specimen collection tube type, Specimen not collected reason, Specimen storage temperature value, Specimen time collection process interval,



✓ Large community effort

✓ Consensual definitions

⚠ Flat list of terms

⚠ No recommendation data structures

Sample knowledge graphs : SPHN network

[nature](#) > [scientific data](#) > [articles](#) > [article](#)

Article | [Open Access](#) | [Published: 10 March 2023](#)

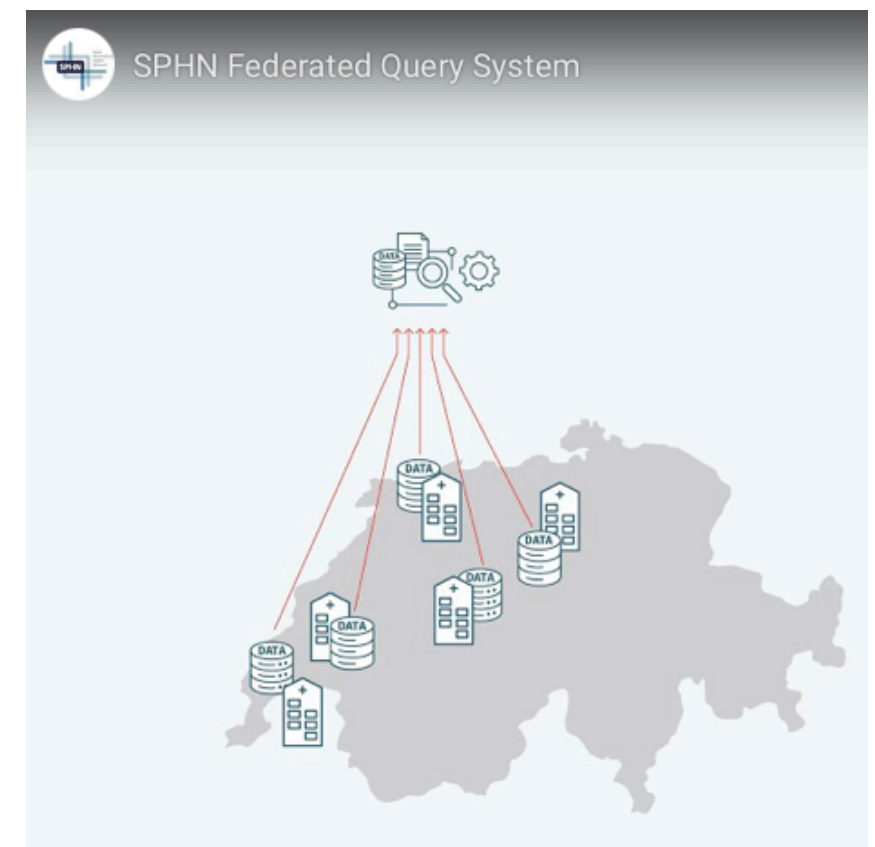
FAIRification of health-related data using semantic web technologies in the Swiss Personalized Health Network

[Vasundra Touré](#), [Philip Krauss](#), [Kristin Gnodtke](#), [Jascha Buchhorn](#), [Deepak Unni](#), [Petar Horki](#), [Jean Louis Raisaro](#), [Katie Kalt](#), [Daniel Teixeira](#), [Katrin Crameri](#) & [Sabine Österle](#) 

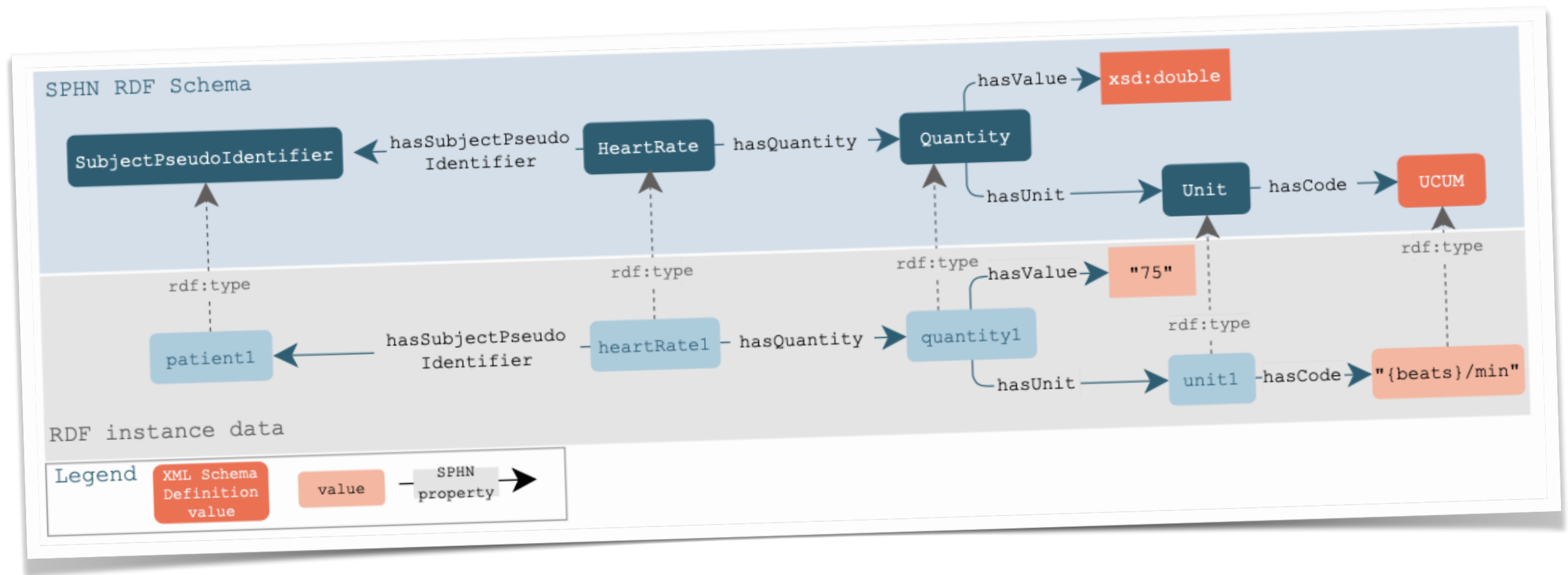
[Scientific Data](#) **10**, Article number: 127 (2023) | [Cite this article](#)

Abstract

The Swiss Personalized Health Network (SPHN) is a government-funded initiative developing federated infrastructures for a responsible and efficient secondary use of health data for research purposes in compliance with the FAIR principles (Findable, Accessible, Interoperable and Reusable). We built a common standard infrastructure with



Sample knowledge graphs : SPHN network



- + Allergy
- + Biobank
- + Consent
- + Radiology diagnosis
- + Genomic variants position
- + ECG

Sample knowledge graphs: Malik et. al. 2020

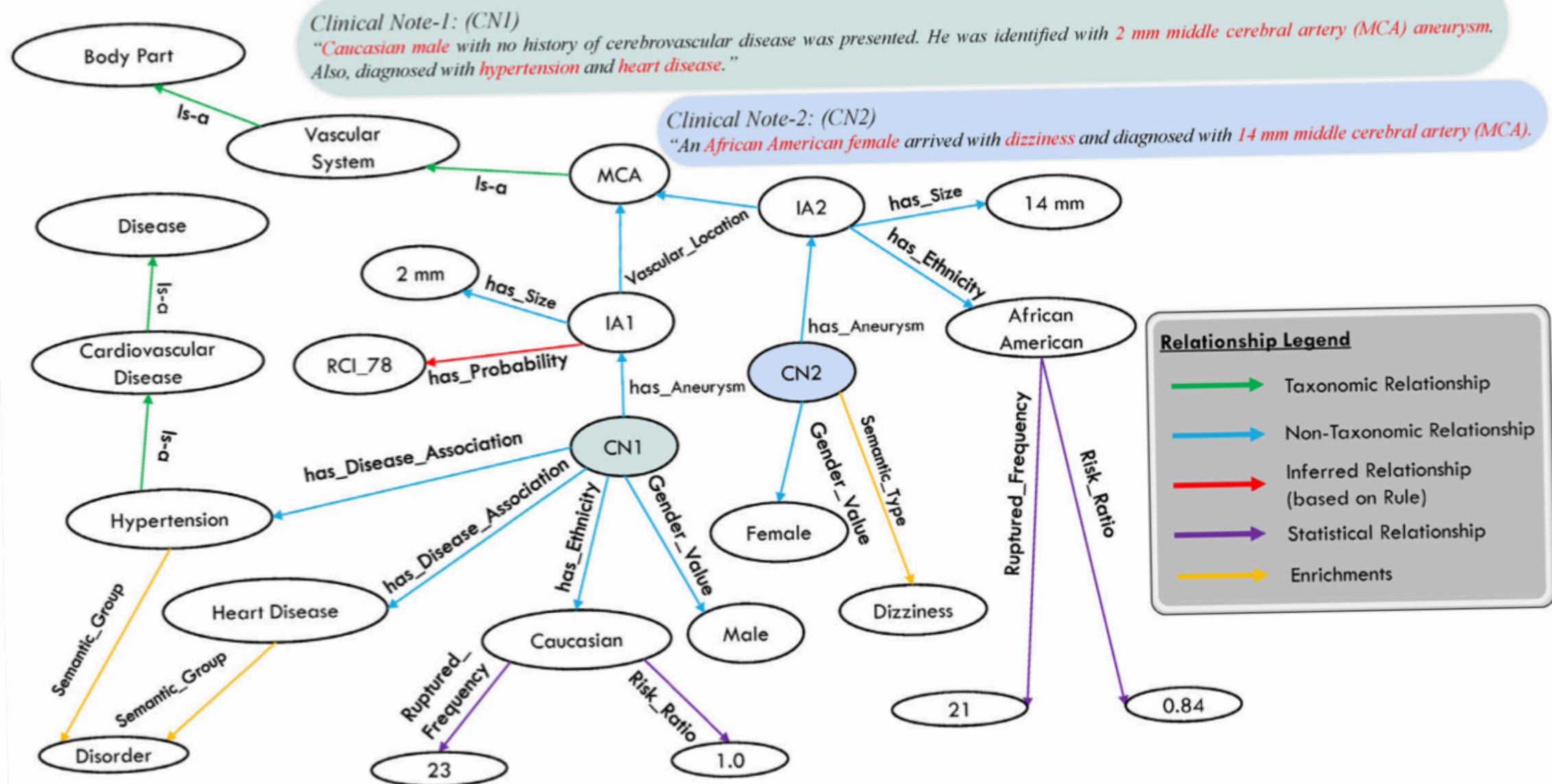
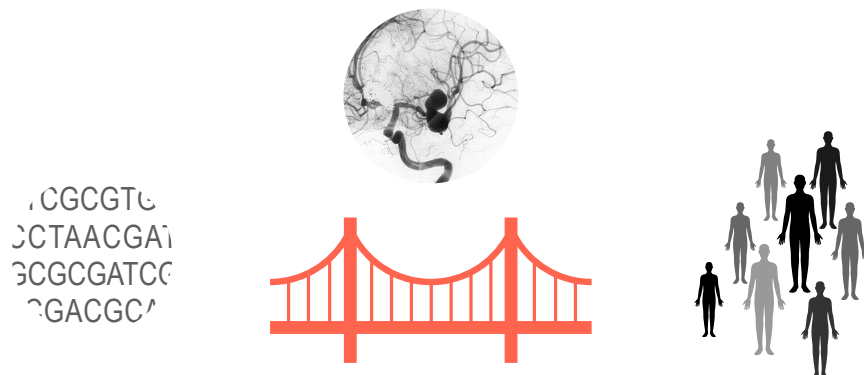


Fig. 3. An excerpt/sub-graph of saccular aneurysm KG of middle cerebral artery defining IA risk factors based on two clinical texts.

Future works

Wrap up & next steps

- ▶ Gather **consensual expert knowledge** with computational **Ontologies**
- ▶ Knowledge Graphs are "**machine-processable**"
- ▶ Link **multi-scale data** with Knowledge Graphs
- ▶ Query **multi-source** data



PEPR Santé numérique

- ▶ Include familial relations to query IA **familial forms**
- ▶ Represent **care pathways and guidelines**
- ▶ Develop a national **data hub** with FAIR intracranial aneurysms data
- ▶ **Share** and enrich data models with **international** initiatives, *but which ones ?*
- ▶ Further develop **predictive approaches** (rupture risk prediction)



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Julie Thompson



ICAN consortium



BiRD facility,
NeurInfo facility

